# Social Security and the Distribution of Effective Retirement Incentives

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**Abstract**

This paper asks how much does Social Security transform a common payroll-tax reform into different effective labor wedges across older workers, and how much do those wedge differences change retirement behavior? Standard policy analysis often treats payroll-tax changes as uniform incentive shifts, but for seniors the relevant object is not the statutory tax rate itself. It is the effective wedge on work after the benefit formula, claiming rules, and the Retirement Earnings Test are taken into account. Exploiting the 2011–2012 U.S. payroll-tax holiday with monthly SIPP data on workers ages 62–70, we find that retirement responses vary systematically with baseline tax exposure, claiming status, and the strength of marginal benefit accrual. Already-claimed workers respond most, consistent with facing a near-pure tax. A dynamic retirement model calibrated to pre-reform moments shows that the Retirement Earnings Test is a major barrier to pre-FRA work, and that benefit linkage sustains late-career employment. The policy incidence of payroll-tax reform for seniors depends on workers' institutional position within Social Security, not just on the statutory rate.

**Keywords:** Social Security, payroll taxes, labor supply, retirement, mortality risk, policy incidence.

**JEL codes:** H55, J14, J22, H31, H24.

# 1 Introduction

Workers in the United States pay Social Security payroll taxes while they work and later receive monthly retirement benefits. At first glance, a change in the payroll-tax rate should therefore change work incentives in the same way for everyone. For older workers, however, the link between today's payroll taxes and tomorrow's benefits differs sharply across people. Workers who have not yet claimed Social Security can still raise their future benefit by working another year, so the payroll tax is partly a payment for something valuable. Workers who have already claimed get little or nothing back — the payroll tax is approximately a pure tax. The progressive benefit formula, the claiming decision, and the Retirement Earnings Test together determine how much of the payroll tax is offset by future benefits. A payroll-tax change therefore does not create a common incentive change across older workers.

This paper asks a quantitative question that is central to Social Security policy but has not been answered directly: how much does Social Security transform a common payroll-tax reform into different effective labor wedges across older workers, and how much do those wedge differences change retirement behavior? The question matters because payroll-tax rate changes are the primary instrument in the solvency debate, and the Office of the Chief Actuary routinely evaluates proposals to raise the Old-Age, Survivors, and Disability Insurance (OASDI) rate to reduce the program's financing shortfall (Social Security Administration, Office of the Chief Actuary, 2025; Goss, 2019). Yet policy debates continue to focus on the statutory tax rate alone. For older workers, this is insufficient – what matters is the effective tax wedge on labor, net of benefit accrual, claiming rules, and the Retirement Earnings Test.

A large literature already shows that Social Security affects retirement decisions, but it has not studied this question directly. One strand identifies retirement responses from cross-sectional differences in workers' own incentive positions, such as accrual rates, option values, or proximity to the earnings-test kink (Coile and Gruber, 2007; French, 2005; Gelber et al., 2022). Another strand studies common statutory reforms with differential exposure across cohorts, such as the increase in the Full Retirement Age (Mastrobuoni, 2009) or the 1977 benefit notch (Krueger and Pischke, 1992). While these papers provide important groundwork, they do not directly address the key policy question: given a uniform payroll-tax change affecting all older workers, how much treatment heterogeneity does Social Security generate, and to what extent does this heterogeneity affect retirement behavior? Without such an answer, policy analysis rely on a single labor-supply elasticity for seniors, even though economically relevant object is a distribution of tax wedges.

We answer that question using the 2011–2012 U.S. payroll-tax holiday, which temporarily lowered the employee OASDI rate from 6.2 to 4.2 percent. The reform is well suited to our setting because the statutory tax cut was common, but its effective size was not. Using monthly SIPP data on workers ages 62–70, we compare workers who entered the reform with different predetermined institutional positions: different baseline exposure to the employee payroll tax, different claiming status, and, among workers who had not yet claimed, different rates at which additional earnings translated into future benefits under the Social Security formula. In this way, the paper quantifies both sides of the missing mapping: how Social Security creates dispersion in effective wedges, and how that dispersion changes retirement responses to a common reform.

Why can the same statutory reform produce different responses even among seniors who all face the same headline tax cut? Two forces matter. First, some workers are already close to retirement, so a small change in incentives moves a relatively large share of them across the exit margin. This is a density channel: the response is larger when more workers are bunched near retirement. Second, some workers have more to gain from staying employed because continued work preserves future flexibility and can raise future Social Security benefits. This is an option-value channel. In our setting, claimed workers are often more exposed to the first force, while unclaimed workers with stronger accrual incentives are more exposed to the second. The paper's empirical and quantitative analysis is designed to measure how these two forces interact with the distribution of effective wedges created by Social Security.

Three findings stand out. First, workers with larger baseline payroll-tax exposure shift retirement timing more during the holiday. Second, workers who had already claimed benefits respond more strongly than otherwise similar workers who had not yet claimed. Third, among workers who had not yet claimed, responses vary systematically with the strength of marginal benefit accrual. The common message is simple: for seniors, retirement responses depend not only on the statutory tax rate, but also on how tightly current payroll taxes are linked to future benefits.

That mechanism is conceptually familiar. When payroll taxes buy future benefits, they are less distortionary than a pure tax. Feldstein and Samwick (1992) showed theoretically that the progressive OASDI benefit formula creates a wide distribution of effective marginal tax rates across workers, depending on where their earnings fall relative to the bend points. Feldstein (2005) formalized the more general point that benefit linkage attenuates the distortion from payroll taxation. Our contribution is to show why this mechanism should be central, not peripheral, in the analysis of payroll-tax reform for seniors. We take the heterogeneity in effec-

tive rates that Feldstein and Samwick (1992) documented computationally and test whether older workers respond to it when a common payroll-tax reform makes those differences bite. The heterogeneity appears on two margins: a discrete margin, whether a worker has already claimed, and a continuous margin, the worker's marginal benefit-accrual rate under the formula. We then use the same effective-wedge idea to connect the reduced-form evidence to a dynamic retirement model for seniors, so that the empirical design and the structural interpretation share a common object.

We interpret these facts with a parsimonious dynamic retirement model for seniors. The model is partial equilibrium and begins at age 62, taking earlier earnings and wealth accumulation as given. In the benchmark theory, claiming status is taken as given, matching the empirical design that compares claimed and unclaimed workers conditional on observed status. The richer quantitative implementation then allows initially unclaimed workers to adjust claim timing in the counterfactuals. The point is not simply to add Social Security to a retirement model. The point is to use the model to measure how much policy incidence changes once the heterogeneity in effective wedges is taken seriously. The model therefore serves two purposes. First, it organizes the reduced-form heterogeneity through a single object, the effective wedge on work. That object makes transparent how claiming status, benefit accrual, survival expectations, and the Retirement Earnings Test each shift the payoff to one more period of work. Second, calibrated to pre-reform employment moments, the model quantifies how much these wedge differences matter for policy incidence. We study a permanent payroll-tax cut, the elimination of the Retirement Earnings Test, and a reform that shuts down post-62 benefit accrual from additional earnings while holding benefit levels fixed.

The main message of the paper is straightforward. For seniors, a payroll-tax change is not a common incentive change. Social Security turns the same statutory reform into different effective wedges across workers, and those wedge differences materially shape retirement behavior. The two counterfactuals speak to different groups and different margins: eliminating the Retirement Earnings Test matters most for low-asset early claimers, while removing post-62 accrual matters most for seniors who have enough resources to delay claiming. Taken together, they show that Social Security shapes late-life labor supply through two separate channels, and each channel matters for a different part of the wealth distribution. The missing ingredient in current policy analysis is therefore not another average retirement elasticity. It is a quantitative account of how Social Security maps one statutory reform into many effective incentives.

### Putting in the Literature

This paper connects to three strands of the literature. First, our paper builds on the insight that benefit linkage attenuates the distortion from payroll taxation (Feldstein, 2005). Feldstein and Samwick (1992) documented computationally that the progressive benefit formula generates a wide cross-sectional distribution of effective marginal tax rates, but they did not estimate behavioral responses to a rate change. Liebman et al. (2009) and Liebman and Luttmer (2015) show that workers respond to Social Security incentives and to their understanding of those incentives, while Gelber et al. (2022) estimate employment responses at the earnings-test kink. A separate literature uses common statutory reforms with differential bite to estimate retirement effects: Krueger and Pischke (1992) exploit the 1977 benefit notch, while Mastrobuoni (2009) and Behaghel and Blau (2012) use the legislated increase in the Full Retirement Age. Those reforms change benefit levels or the claiming incentive. Our design is different: the treatment is a payroll-tax rate change, the heterogeneity comes from the nonlinear mapping of earnings to benefits, and the outcome is continued work versus retirement. We do not claim a new incentive mechanism; rather, we use a common rate change to test whether the effective-rate heterogeneity documented by Feldstein and Samwick (1992) produces differential retirement responses.

Second, this paper contributes to the literature on structural retirement models with Social Security incentives. A large literature models retirement as a dynamic decision shaped by Social Security rules, option values, health, and wealth (Stock and Wise, 1990; Rust and Phelan, 1997; French, 2005; French and Jones, 2011; Coile and Gruber, 2007; Coile, 2023; Blundell et al., 2016). Our paper takes a different perspective on that question. Instead of using the model mainly to measure average retirement effects of Social Security, we use it to understand why the same payroll-tax reform leads different seniors to respond differently. The model is therefore organized around the same effective-wedge idea as the reduced-form evidence. Its role is to interpret heterogeneous incidence and to separate the channels through which benefit linkage and the Retirement Earnings Test affect retirement.

Third, we contribute to the literature on mortality heterogeneity and tax wedges. The broader lifecycle literature shows that mortality risk changes labor-supply and retirement incentives (Yaari, 1965; De Nardi et al., 2006, 2016), and work on subjective survival expectations shows that horizons help explain retirement behavior (Hurd and McGarry, 1995, 2002; Hurd et al., 2004). At the same time, macro labor-supply work often summarizes tax incentives with a single wedge (Prescott, 2004; Ohanian et al., 2008; Rogerson and Wallenius, 2009; Chetty et al., 2013; Laun and Wallenius, 2016). Our paper links these ideas in a narrow way:

survival matters here because it changes the value of benefit accrual, and the relevant policy object is not one senior wedge but the distribution of effective wedges created by Social Security institutions.

### Roadmap

The remainder of the paper proceeds as follows. Section 2 develops a simple theory of how survival and Social Security incentives shape retirement behavior. Section 3 describes the data and institutional background. Section 4 presents the motivating evidence on senior labor supply around the payroll-tax holiday. Sections 5–7 develop, parameterize, and calibrate the dynamic retirement model for seniors. Section 8 reports the counterfactual exercises. Section 9 concludes.

## 2 A Simple Model of Retirement

### 2.1 Environment

**Preferences and Survival** Consider a worker who is currently age $a_0$ and is deciding how much to work and consume over the rest of her life. She faces uncertainty about how long she will live. We capture this with a survival function $s(t \mid \pi)$, which gives the probability that a worker of mortality type $\pi$ survives at least $t$ more years beyond age $a_0$. By construction, $s(0 \mid \pi) = 1$ (you are alive today with certainty) and $s(T \mid \pi) = 0$ (no one lives beyond some maximum horizon $T$). Throughout, time $t$ is measured in years after $a_0$. The force of mortality—the instantaneous risk of dying at each moment, conditional on having survived to that point—is

$$\delta(t \mid \pi) \;=\; -\frac{\dot{s}(t \mid \pi)}{s(t \mid \pi)},$$

and the expected remaining lifetime is simply the area under the survival curve: $H(\pi) \equiv \int_0^T s(t \mid \pi)\, dt$.

The worker chooses consumption $c_t$ and hours of work $h_t$ to maximize expected lifetime utility:

$$\max_{\{c_t,\, h_t\}} \int_0^T e^{-\rho t}\, s(t \mid \pi) \Big[ u(c_t) - v(h_t) - F \cdot \mathbf{1}[h_t > 0] \Big]\, dt. \tag{1}$$

The three terms inside the brackets have intuitive roles. The first, $u(c_t)$, is the pleasure from consumption (increasing and concave, so each additional dollar brings less satisfaction). The second, $v(h_t)$, is the discomfort of working (increasing and convex, so each additional hour

feels harder). The third, $F \cdot \mathbf{1}[h_t > 0]$, is a fixed cost incurred whenever the worker participates at all—think of commuting, maintaining professional credentials, or the rigidity of a work schedule. Future utility is discounted both because people are impatient (at rate $\rho > 0$) and because they may not be alive to enjoy it (captured by $s(t \mid \pi)$).

**Budget Constraint and the Definition of $\lambda$** The worker's financial wealth $A_t$ grows according to

$$\dot{A}_t \;=\; rA_t + w_t\, h_t\,(1 - \tau + \lambda\,\mu_t) - c_t. \tag{2}$$

Reading this left to right: wealth increases with interest earned on existing assets ($rA_t$) and with labor income, and decreases with consumption ($c_t$). The interesting part is the labor-income term. The worker earns a pre-tax wage $w_t$ for each hour $h_t$, but not all of her payroll tax $\tau$ is truly lost. The term $\lambda\,\mu_t$ captures how much comes back as future Social Security benefits. Here $\lambda \geq 0$ is the marginal annual benefit accrual per dollar of earnings—it depends on where the worker sits relative to the bend points in the Social Security formula and is independent of the tax rate $\tau$. The factor $\mu_t$ is the annuity factor: the present value of receiving \$1 per year in future benefits, weighted by the worker's survival probability. Together, $\lambda\,\mu_t$ measures how much each dollar of earnings today is effectively "refunded" through higher future benefits, so the true cost of the payroll tax is not $\tau$ but $\tau - \lambda\,\mu_t$.

**The Annuity Factor and Its Mortality Dependence** How much is a future benefit stream worth today? That depends on how long you expect to live to collect it. The annuity factor captures exactly this idea. For a worker of type $\pi$, it asks: what is the present value of receiving \$1 per year for the rest of your life, given your survival prospects $s(\cdot|\pi)$?

All integrals run over $t \in [0, T]$, where $T$ is the number of years remaining after age $a_0$. The annuity factor is:

$$\mu_0(\pi) \;=\; \int_0^T e^{-rt}\, s(t \mid \pi)\, dt. \tag{3}$$

Each future dollar is discounted by the interest rate ($e^{-rt}$) and weighted by the probability of being alive to receive it ($s(t \mid \pi)$). A worker who expects to live longer has a higher $\mu_0$, because more of those future dollars are likely to be collected.

The parameter $\lambda$ captures how much an extra dollar of earnings today raises your annual Social Security benefit—this depends on where you sit relative to the bend points in the benefit formula. The product $\lambda\mu_0(\pi)$ then tells you the total present value of the extra benefits you earn by working one more dollar. Notice that the units work out: $\lambda$ is in (\$/year per \$ of

earnings) and $\mu_0$ is in (years), so $\lambda\mu_0$ is dimensionless. This is why the effective price of work, $1 - \tau + \lambda\mu_0(\pi)$, is a clean, unit-free number that we can compare across workers.

**Mortality ordering and monotonicity.** To compare workers with different survival prospects, we rank them in the natural way: type $\pi'$ has better survival than type $\pi$ if $s(t \mid \pi') \geq s(t \mid \pi)$ at every age, with a strict advantage somewhere. Under this ordering, both life expectancy,

$$H(\pi) = \int_0^T s(t \mid \pi)\, dt,$$

and the annuity factor,

$$\mu_0(\pi) = \int_0^T e^{-rt} s(t \mid \pi)\, dt,$$

are strictly increasing in $\pi$. The logic is straightforward: if you are more likely to be alive at every future date, both the expected years of life and the present value of a benefit stream must be higher. Consequently, workers who expect to live longer place a higher value on the future benefits that come from working today.

## 2.2 Optimality conditions

The current-value Hamiltonian with costate $\xi_t$ is

$$\mathcal{H} \;=\; u(c_t) - v(h_t) - F \cdot \mathbf{1}[h_t > 0] + \xi_t[rA_t + w_t h_t(1 - \tau + \lambda\mu_t) - c_t]\,.$$

Discounting by survival probability makes the effective discount rate $\rho + \delta(t \mid \pi)$, so the costate satisfies

$$\dot{\xi}_t \;=\; \xi_t(\rho + \delta(t \mid \pi) - r)\,. \tag{4}$$

Conditional on participation ($h_t > 0$), the fixed cost $F$ does not depend on $h$ and drops out of the first-order condition:

$$v'(h_t) \;=\; \xi_t\, w_t(1 - \tau + \lambda\mu_t)\,. \tag{5}$$

Evaluating at $t = 0$ for each worker type:

$$v'(h_{\mathrm{c}}) = \xi_{\mathrm{c}}\, w_0\,(1 - \tau) \qquad \text{(Claimed, approximation } \lambda \approx 0) \tag{6}$$

$$v'(h_{\mathrm{u}}) = \xi_{\mathrm{u}}\, w_0\,(1 - \tau + \lambda_0\mu_0) \qquad \text{(Unclaimed, } \lambda_0 > 0) \tag{7}$$

where $\lambda_0\mu_0 \equiv \lambda\,\mu_0(\pi)$ with $\mu_0(\pi)$ as in (3).

$$\phi \;\equiv\; \frac{v'(h)}{h\,v''(h)} \quad\Longleftrightarrow\quad \frac{v''(h)}{v'(h)} = \frac{1}{\phi h}. \tag{8}$$

The Frisch elasticity $\phi > 0$ is a preference parameter. It does not depend on the tax code, the benefit formula, or the mortality type.

The fixed cost $F$ introduces an extensive margin. Conditional on $h_t > 0$, the first-order condition (5) is unchanged because $F$ does not depend on $h$. The agent works at date $t$ if and only if the flow surplus from interior-optimal hours exceeds the fixed cost:

$$\xi_t\, w_t(1-\tau+\lambda\mu_t)\, h_t^* - v(h_t^*) \;\geq\; F, \tag{9}$$

where $h_t^*$ solves (5).

**Closed form under power disutility.** With $v(h) = h^{1+1/\phi}/(1+1/\phi)$, the interior optimum is $h_t^* = [\xi_t\, w_t(1-\tau+\lambda\mu_t)]^{\phi}$ and the surplus simplifies to

$$\frac{[\xi_t\, w_t(1-\tau+\lambda\mu_t)]^{1+\phi}}{1+\phi}.$$

The participation condition is therefore

$$h_t = 0 \quad\Longleftrightarrow\quad \xi_t\, w_t(1-\tau+\lambda\mu_t) \;<\; \big[(1+\phi)\,F\big]^{\frac{1}{1+\phi}}. \tag{10}$$

The right-hand side defines an effective reservation wage. If the shadow value of an hour of work falls below this threshold, the agent does not participate. Corner solutions are more likely when the fixed cost $F$ is large, when the agent is wealthy (low costate $\xi_t$), when the market wage $w_t$ is low, or when the net-of-tax-plus-accrual rate $(1-\tau+\lambda\mu_t)$ is low.

So far, this is only an individual decision rule. It tells us when one worker chooses to work and when one worker chooses not to work. But our paper is about groups of workers. We want to know why the same payroll-tax reform changes employment more for some groups of seniors than for others. Hence, we need one more ingredient: a distribution of participation thresholds within each group.

Let $g$ index a group of workers, such as claimed workers, unclaimed workers, or high-survival unclaimed workers. For each worker $i$ in group $g$, define the *effective reservation wage*

$$\omega_i \;\equiv\; \frac{\big[(1+\phi)F_i\big]^{1/(1+\phi)}}{\xi_i w_i},$$

which is the minimum effective return to work that makes participation worthwhile. This object collects all the worker-specific reasons that make work harder to sustain: fixed costs, the marginal value of wealth, and wages. Worker $i$ in group $g$ participates if and only if

$$1 - \tau + \lambda_g \mu_g \;\geq\; \omega_i.$$

Now let $F_g^\omega$ denote the cumulative distribution function of reservation effective wages in group $g$. Then the employment rate in group $g$ is

$$N_g(\tau) \;=\; F_g^\omega\big(1 - \tau + \lambda_g \mu_g\big). \tag{11}$$

This is the bridge from theory to the data. It says that a common payroll-tax reform changes employment by moving each group left or right along its own distribution of reservation wages. Differentiating (11) gives

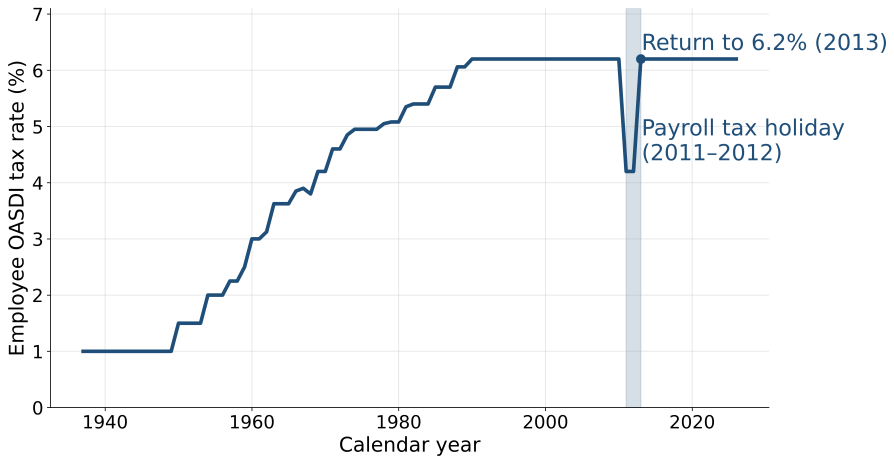
$$\frac{\partial N_g(\tau)}{\partial \tau} \;=\; -\, f_g^\omega\big(1 - \tau + \lambda_g \mu_g\big), \tag{12}$$

where $f_g^\omega$ is the corresponding density. The expression shows that a common statutory tax change need not produce a common employment response. Groups differ in $\lambda_g \mu_g$ and therefore start from different effective prices of work. At the same time, even groups facing identical effective price changes will respond differently if the mass of workers near the participation margin—captured by $f_g^\omega$ at the relevant threshold—differs across them.

Two distributions are therefore central to the analysis. First, the distribution of *effective wedges*, determined by claiming status, benefit accrual, and survival expectations—captures how Social Security differentially changes the price of work for each senior. Second, the distribution of *effective reservation wages* within each group captures how many workers are near enough to the margin to act on those price changes. The quantitative question at the heart of the paper is how much Social Security widens the distribution of effective wedges under a common reform, and how much that widening matters for employment and retirement.

Answering this question requires moving from the model's objects to observable institutions. The payroll-tax holiday, combined with differences in claiming status, benefit accrual, and the Retirement Earnings Test, generates measurable variation in effective incentives across older workers. The reduced-form analysis then tests whether workers facing larger effective wedge changes do, in fact, adjust retirement behavior by proportionally larger amounts.

Figure 1: Employee OASDI Payroll-Tax Rate Path, 2010–2013




*Notes:* The employee OASDI rate is 6.2% in 2010, 4.2% in 2011–2012, and 6.2% in 2013.

# 3 Data and Institutional Background

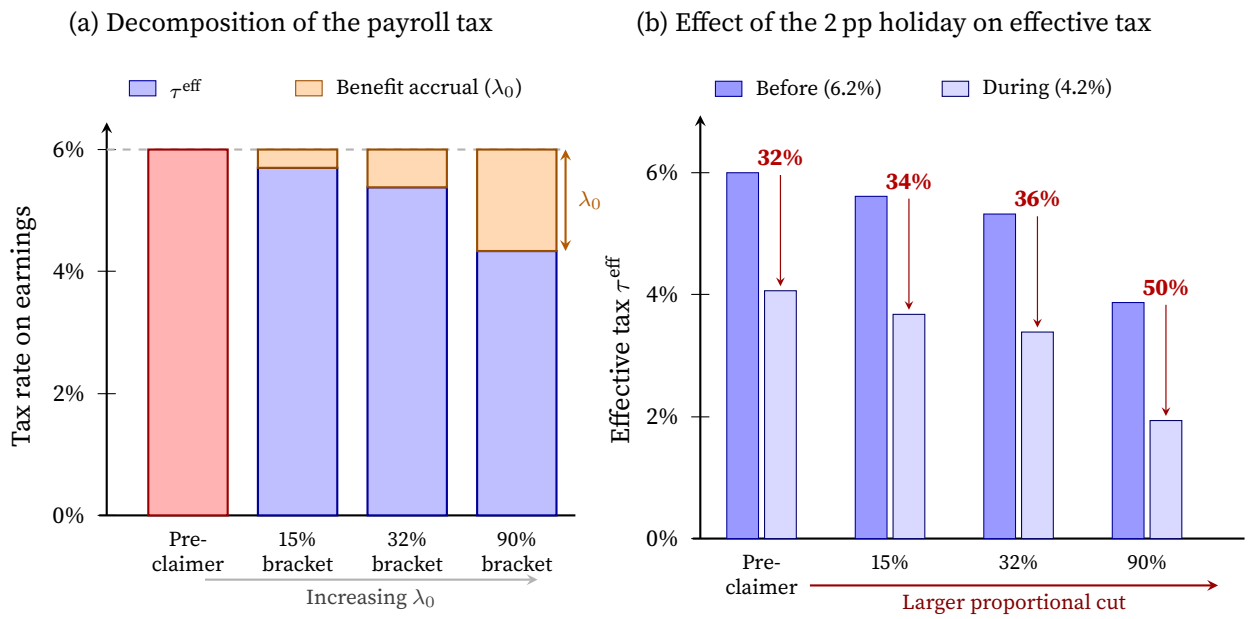
## 3.1 Institutional Background

**Payroll-Tax Policy Timing** Our empirical setting is the temporary federal payroll-tax holiday in the United States. For employees, the Old-Age, Survivors, and Disability Insurance (OASDI) payroll-tax rate on covered earnings was 6.2% in 2010, fell by 2 percentage points to 4.2% in January 2011, remained at that level through December 2012, and then returned to 6.2% in January 2013. The policy therefore generates two sharp and well-defined time breaks: an expansionary shock at the start of 2011 and a reversal at the start of 2013.

Because the reform was federal, it did not roll out differentially across states or sectors in our analysis sample; all covered workers faced the same statutory rate path at the same calendar dates. This institutional feature is important for interpretation: identifying variation in our design comes from differential responses across pre-policy worker types within common national policy-time shocks, rather than from geographic policy adoption differences.

**Claiming Status and Incentives** At ages 62–70, labor supply decisions are tightly linked to Social Security claiming stage. Individuals who have not yet claimed remain on a margin where continued work and timing of claiming jointly determine future retirement-income flows, while individuals who already claimed are typically on a different effective-return margin for additional work. In practice, this creates systematic differences in how a temporary change in current take-home pay translates into labor-supply incentives near retirement.

Figure 2: Institutional Heterogeneity in the Effective Payroll Tax and the Holiday's Differential Impact




*Notes:* Illustrative example for an unclaimed worker at age 64. Panel (a): the statutory OASDI rate ($\tau = 6.2\%$) decomposes into an effective tax ($\tau^{\text{eff}} = \tau - \lambda_0$, blue) and a benefit-accrual offset ($\lambda_0$, orange). Pre-claimers have $\lambda_0 \approx 0$; unclaimed workers' $\lambda_0$ varies with position on the PIA schedule. Panel (b): the 2 pp holiday reduces $\tau$ from 6.2% to 4.2% for all workers, but the proportional reduction in $\tau^{\text{eff}}$ is larger for high-$\lambda_0$ workers (50% for the 90% bracket vs. 32% for pre-claimers) because their effective tax is smaller to begin with. PIA brackets refer to the marginal replacement rate at the worker's 2010 AIME position: 90% below the first bend point, 32% between bend points, and 15% above the second.

For this reason, pre-policy claiming status is a central institutional heterogeneity dimension in our framework. It separates workers by their position in the Social Security lifecycle before the payroll-tax change occurs, allowing us to compare policy responses across groups that plausibly face different effective incentives to continue working.

The distinction is easiest to see through the effective tax wedge. For an unclaimed worker, part of the statutory payroll tax is offset by future benefit accrual, and the size of that offset varies mechanically with position on the benefit formula. For a pre-claimer, that offset is essentially absent, so the same 2 percentage-point holiday produces a different proportional change in the effective tax wedge across workers.

**Horizon Heterogeneity** The second institutional margin is expected horizon. A worker with higher mortality risk has a shorter horizon over which future payoffs from continued work can be realized, while a worker with lower mortality risk has a longer horizon and can value those future payoffs more. This horizon channel affects both the salience of short-run net-

of-tax wage changes and the value of future-oriented retirement-income consequences from continued labor-force attachment.

Accordingly, the same payroll-tax shock can induce different labor-supply responses even within the same claiming group, depending on survival type. In the empirical analysis, survival is therefore used as a supporting heterogeneity dimension: it helps us compare workers at similar points in the life cycle and lets the claiming-status response vary with horizon length.

**Predetermined Heterogeneity in SIPP** Our SIPP analysis uses four sources of heterogeneity that are fixed before the holiday begins. First, baseline payroll-tax exposure is measured from predetermined 2010 covered earnings and determines the dollar size of the 2 percentage-point tax cut. Second, pre-policy claiming status, denoted $\text{PreClaimer}_i$, is measured from the worker's latest observed month in 2010. This separates workers for whom payroll taxes are close to a pure tax from workers for whom those taxes are still partly offset by future benefit accrual. Third, baseline survival, denoted $S_i$, is a transported survival proxy constructed from 2010 covariates and then held fixed throughout estimation. Fourth, among unclaimed workers with observed earnings, we construct baseline $\lambda_{0,i}$, the mechanical marginal increase in future Social Security benefits generated by one more year of covered earnings.

By construction, all four objects are predetermined with respect to the 2011–2012 holiday. This limits reverse-causality concerns, because the policy cannot change a worker's baseline exposure, pre-holiday claiming status, baseline survival proxy, or baseline benefit-accrual measure.

### 3.2 Empirical Strategy

The paper does not rely on a single regression. Instead, the empirical section uses three related difference-in-differences comparisons built around the same national policy timing. In each case, the time-series variation comes from the payroll-tax holiday itself, and the cross-sectional variation comes from predetermined worker characteristics measured before the holiday. Because all covered workers faced the same statutory rate cut at the same time, identification comes from asking whether workers with different baseline exposure changed by different amounts.

The common treatment formula is

$$x_{it} = \text{Post}_t \times s_i,$$

where $\text{Post}_t$ indicates the holiday period and $s_i$ is a predetermined exposure measure. This notation is useful because it makes the design transparent. The national policy shock is the same for everyone and is absorbed by year-month fixed effects. The exposure measure $s_i$ is fixed before the holiday and is absorbed in levels by person fixed effects. What remains is the interaction between the two: whether workers with larger baseline exposure change retirement or employment more during the holiday.

The main reduced-form evidence focuses on the cut episode, comparing the holiday years 2011–2012 with the pre-holiday year 2010. The three designs are straightforward.

1. **Tax-exposure difference-in-differences.** Here $s_i$ is baseline payroll-tax exposure. We compare workers with high versus low exposure to ask whether a larger dollar tax cut produces a larger change in retirement timing.

2. **Claiming-status difference-in-differences.** Here $s_i$ is pre-policy claiming status. We compare pre-claimers and non-claimers to ask whether the same statutory reform has a larger effect when payroll taxes are less tied to future benefits. In this specification the main estimand is still the coefficient on $\text{Post}_t \times \text{PreClaimer}_i$. We also include $\text{Post}_t \times \text{Survival}_i$ and $\text{Post}_t \times \text{PreClaimer}_i \times \text{Survival}_i$ so the claiming-status response can vary with horizon length and so the main coefficient is interpreted at average survival.

3. **Continuous-treatment difference-in-differences within the unclaimed sample.** Here $s_i$ is baseline $\lambda_{0,i}$. We restrict attention to unclaimed workers and interact the holiday period with baseline $\lambda_0$ to ask whether the holiday effect rises smoothly with the strength of marginal benefit accrual itself.

This sequence matches the logic of the paper. The first design provides broad reduced-form evidence that retirement behavior moved around the holiday and that the movement was larger for workers with greater baseline exposure. The second shows that the response differs across workers on opposite sides of the benefit-linkage margin. The third goes one step deeper and asks whether the response varies continuously with the model's central state variable, $\lambda_0$.

This is best understood as a common-shock exposure design. In that respect, it is closer in spirit to Finkelstein (2007), who studies how the common introduction of Medicare had different effects across markets with different baseline exposure, than to staggered-adoption two-way fixed-effects designs. The main concern is therefore not forbidden comparisons generated by different treatment timing, because treatment timing is common in our setting. At

the same time, we do not treat the holiday as randomized. Identification still relies on a conditional parallel-trends restriction for these exposure-based comparisons.

The common identifying assumption is a conditional parallel-trends restriction for each comparison. Absent the holiday, the relevant groups should not have experienced systematically different changes in retirement or employment over the 2010–2012 window, once person fixed effects absorb permanent differences across workers and year-month fixed effects absorb shocks common to all workers in a given month. Because the comparison variables are measured before the holiday, this design also avoids mechanical reverse causality from policy exposure to treatment assignment. The empirical section probes the remaining concerns using timing patterns, placebo comparisons, and alternative specifications, and Section B reports the corresponding monthly timing diagnostics.

This section defines the SIPP sample, variables, and construction of mortality heterogeneity used in the main empirical analysis.

### 3.3 Data Source and Unit of Observation

Our main data are from the 2008 panel of the Survey of Income and Program Participation (SIPP). We use the monthly core records because the payroll-tax holiday begins at a precise calendar boundary, January 2011. In the main empirical analysis, the window starts in January 2010 and ends in December 2012. This gives us 12 clean pre-holiday months followed by 24 holiday months. The baseline observation unit is a person-month pair $(i, t)$.

### 3.4 Sample Construction

We begin from all SIPP monthly records and restrict the analysis to respondents ages 62–70, the age range where both labor-supply adjustment and Social Security claiming behavior are economically relevant for our setting. For the main reduced-form estimates, we keep calendar years 2010–2012. We require non-missing labor outcomes and valid claiming assignment in each person-month used for estimation. In specifications that interact treatment with survival heterogeneity, we further require a non-missing baseline transported survival measure constructed from 2010 information, so heterogeneity is fixed before policy exposure.

The resulting panel is unbalanced at the month level because respondents can rotate out of interview months or have missing items in specific months. The broad age-62–70 sample over January 2010 through December 2012 contains 270,269 person-months from 12,036 individuals. The estimating sample is smaller in particular specifications when we impose additional baseline-earnings, claiming-status, survival, or risk-set restrictions.

Table 1: Empirical Designs in the SIPP Analysis

| **Design element** | **Object** | **Role in the empirical analysis** |
|---|---|---|
| *Panel A: Common time variation* | | |
| Policy timing | 2010 vs. 2011–2012 | Common before-after variation from the federal payroll-tax holiday; all covered workers face the same statutory rate cut at the same calendar dates |
| *Panel B: Predetermined cross-sectional variation* | | |
| Baseline exposure | 2010 payroll-tax bite | Determines the dollar size of the holiday for Fact 1: high-exposure workers receive a larger tax cut |
| Claiming status | Pre-claimer vs. non-claimer | Captures whether payroll taxes are close to a pure tax or partly offset by future benefit accrual in Fact 2 |
| Baseline survival | Transported survival proxy | Supporting heterogeneity dimension; allows the claiming-status response to vary with horizon length |
| $\lambda_0$ among unclaimed | Marginal benefit accrual | Continuous measure of benefit linkage used in Fact 3 to test whether the holiday response rises with accrual strength |
| *Panel C: Main estimands* | | |
| Fact 1 | Exposure DiD | Over-time change in retirement hazard, high- vs. low-exposure workers |
| Fact 2 | Claiming-status DiD | Over-time change, pre-claimers vs. non-claimers, evaluated at average survival |
| Fact 3 | Continuous-treatment DiD | Slope of the holiday effect w.r.t. baseline $\lambda_0$ within the unclaimed sample |

*Notes:* The empirical section uses a sequence of related exposure-based difference-in-differences designs rather than a single headline triple-difference regression. The survival interaction in the claiming-status specification allows flexible heterogeneity, but the main Fact 2 coefficient remains a difference-in-differences estimate for workers with average survival.

### 3.5 Key Variables

**Outcomes.** Our primary outcome is monthly employment status. We set

$$\text{Employed}_{it} = \mathbf{1}\{\text{employed}_{it}\},$$

so the indicator equals one for respondents reporting that they are employed (either at work or with a job) and zero otherwise.

Our secondary (corroborative) labor-supply outcome is monthly hours. Let $h_{it}^{w}$ denote mapped weekly hours from the weekly-hours response category, and let $n_{it}$ denote reported weeks worked in the month. We define

$$\text{Hours}_{it} = \begin{cases} 0 & \text{if not employed,} \\ h_{it}^{w} \times n_{it} & \text{if employed,} \end{cases}$$

where $h_{it}^{w}$ maps the reported weekly-hours category to a weekly-hour point.[1] For descriptive dynamics we also use a 3-month moving average of monthly hours. Because hours are recovered from categorical bins and are more exposed to seam-related measurement noise, we treat hours estimates as corroborative rather than primary.

**Income measures and units.** Monthly total income is measured by `TPMSUM1`, and monthly Social Security income is measured by `THSOCSEC`. Both are recorded in nominal U.S. dollars for the SIPP reference month. In interpretation, `THSOCSEC` is a household-level Social Security income aggregate rather than a pure own-benefit flow, so non-claimers can still show positive values when another household member (for example, a spouse) receives benefits.

**Policy timing indicators.** Policy exposure is encoded with calendar-time indicators. For the payroll-tax holiday period,

$$\text{Post}_t = \mathbf{1}\{\text{year} \in \{2011, 2012\}\}.$$

Thus the main estimation window is January 2010 through December 2012, with $\text{Post}_t = 0$ in 2010 and $\text{Post}_t = 1$ in 2011–2012.

**Pre-policy claiming status.** $\text{PreClaimer}_i$ equals one if respondent $i$ reports receiving own Social Security retirement benefits in the latest observed month of 2010, and zero otherwise.

[1] In the SIPP 2008 recode, the weekly-hours categories are: $0 =$ none, $1 =$ all weeks 35+ hours, $2 =$ all weeks 1–34 hours, $3$–$4 =$ a mix of 35+ and 1–34 hours, $5 =$ some weeks 35+ hours and otherwise 0, and $6 =$ some weeks 1–34 hours and otherwise 0. We convert these categories to continuous weekly-hour points as $0 \mapsto 0$, $\{1, 5\} \mapsto 40$, $\{2, 6\} \mapsto 17.5$, and $\{3, 4\} \mapsto 28.75$. The value 17.5 is the midpoint of the 1–34 range, and 28.75 is the midpoint between 17.5 and 40 for mixed-hour categories. Monthly hours are then this mapped weekly value times the reported number of weeks worked in the month.

This status is fixed before policy exposure and aligns with the January 2010 start of the monthly analysis window.

**Survival heterogeneity proxy in SIPP.** SIPP does not include linked mortality outcomes over the policy window, so we construct $S_i$ via external transport from HRS in three steps. The construction follows two related literatures: mortality-risk indexing in older-adult/HRS settings (Lee et al., 2006; Cruz et al., 2013) and transport of source-sample prediction models to target populations with covariates but no observed outcomes (Steingrimsson et al., 2023).

First, in RAND HRS we estimate a wave-to-wave mortality model on ages 59–72 (waves 1–14), keeping respondents interviewed at wave $w$ with known alive/dead status at $w+1$. With $D_{i,w+1} = 1$ if next-wave status is death, we estimate a weighted probit:

$$\Pr(D_{i,w+1} = 1 \mid X_{iw}) = \Phi\big(\alpha + \beta_1 \text{Age}_{iw} + \beta_2 \text{Age}_{iw}^2 + \beta_3 \text{Female}_i + \beta_4 \text{Disabled}_{iw} + \beta_5 \text{SSI}_{iw}\big)\,.$$

All coefficients are estimated in HRS and then fixed before SIPP analysis.

Second, we map harmonized SIPP covariates (age, sex, disability status, and SSI receipt) into $X_{it}$ and compute month-level transported mortality over the model horizon (2 years):

$$\widehat{m}_{it}^{(2)} = \Phi(\widehat{\alpha} + X_{it}'\widehat{\beta}).$$

We convert to a 10-year horizon using a constant-hazard mapping:

$$\widehat{h}_{it} = -\frac{\log(1 - \widehat{m}_{it}^{(2)})}{2}, \qquad \widehat{m}_{it}^{(10)} = 1 - \exp(-10\widehat{h}_{it}), \qquad \widehat{s}_{it}^{(10)} = 1 - \widehat{m}_{it}^{(10)}.$$

Third, to keep heterogeneity assignment predetermined, we collapse to person-level baseline risk using 2010 months only:

$$\overline{m}_{i,2010}^{(10)} = \frac{1}{N_{i,2010}} \sum_{t \in 2010} \widehat{m}_{it}^{(10)}, \qquad S_i \equiv \overline{s}_{i,2010}^{(10)} = 1 - \overline{m}_{i,2010}^{(10)}.$$

This baseline $S_i$ is then merged to all analysis months and centered in the regressions. In grouped analyses, high/low mortality groups are defined by the top/bottom quartiles of $\overline{m}_{i,2010}^{(10)}$ ($\geq q_{0.75}$ and $\leq q_{0.25}$, respectively).

Relative to the common HRS approach of using respondents' own subjective survival expectations measured in the same sample (Hurd and McGarry, 1995, 2002; Hurd et al., 2004), our design differs in two ways: (i) mortality risk is generated out-of-sample (estimated in HRS,

Table 2: SIPP Summary Statistics for Analysis Variables

| Variable | Full sample | Pre-reform (2010–2010) | Policy window (2011–2012) |
|---|---|---|---|
| Age (years) | 65.6 (2.6) | 65.6 (2.6) | 65.6 (2.6) |
| Employed indicator | 0.353 (0.478) | 0.352 (0.477) | 0.353 (0.478) |
| Monthly earnings (TPEARN) | 1,255 (3,305) | 1,222 (3,259) | 1,272 (3,328) |
| Annualized earnings (12×TPEARN) | 15,058 (39,656) | 14,666 (39,103) | 15,259 (39,935) |
| Earnings / taxable maximum | 0.140 (0.368) | 0.137 (0.366) | 0.141 (0.368) |
| Near-cap indicator (0.85–1.15 of max) | 0.018 (0.131) | 0.017 (0.130) | 0.018 (0.132) |
| SS claiming indicator (ESSSELF=1) | 0.702 (0.457) | 0.710 (0.454) | 0.698 (0.459) |
| SSI receipt indicator (ESTSSI=1) | 0.204 (0.403) | 0.209 (0.407) | 0.201 (0.401) |
| Disability indicator (EDISABL=1) | 0.284 (0.451) | 0.281 (0.449) | 0.285 (0.452) |
| Mortality proxy (10y, HRS transport) | 0.132 (0.108) | 0.131 (0.107) | 0.133 (0.108) |
| Baseline mortality proxy (2010 person mean) | 0.125 (0.099) | 0.132 (0.102) | 0.121 (0.097) |
| High-mortality group (top quartile, 2010) | 0.265 (0.441) | 0.279 (0.449) | 0.258 (0.438) |
| Person-month observations | 270,269 | 91,396 | 178,873 |
| Unique individuals | 12,036 | 9,364 | 10,587 |

Notes: Entries are mean (standard deviation). Pre-holiday is January–December 2010; holiday window is January 2011–December 2012. Employment indicator equals one if RMESR is 1 or 2. Near-cap indicator equals one if annualized earnings are between 85% and 115% of the year-specific OASDI taxable maximum.

applied in SIPP), and (ii) because the target SIPP panel does not observe mortality over the policy window, we do not re-estimate or re-calibrate on SIPP outcomes and instead fix heterogeneity at the 2010 baseline.

### 3.6 Descriptive Statistics

Table 2 reports summary statistics for the SIPP analysis variables, including the transported mortality/survival measures and baseline high-mortality assignment.

Several features of Table 2 are directly relevant for the empirical design. First, the age profile is stable across windows (mean 65.6 in 2010 and 65.6 in 2011–2012), and baseline employment is also very similar (0.352 in 2010 versus 0.353 in the holiday window). That is the pattern we want for a monthly fixed-effects design built around a common policy break rather than around large changes in sample composition. The analysis sample is also large: 270,269 person-months from 12,036 individuals overall, with 91,396 observations in 2010 and 178,873 in 2011–2012.

Second, the main heterogeneity dimensions have substantial support in this window. The Social Security claiming indicator is high but not degenerate (0.710 in 2010 and 0.698 in 2011–2012), leaving meaningful claimer/non-claimer variation. The transported mortality proxy is similarly stable over time (0.131 in 2010 and 0.133 in 2011–2012), and the high-mortality group remains close to one quarter of the sample by construction. Finally, near-cap observations are

a small share of the full sample (about 1.8%), so the main estimates reflect broad late-career labor-supply responses rather than behavior concentrated only near the taxable maximum.

# 4 Motivating Evidence on Senior Labor Supply

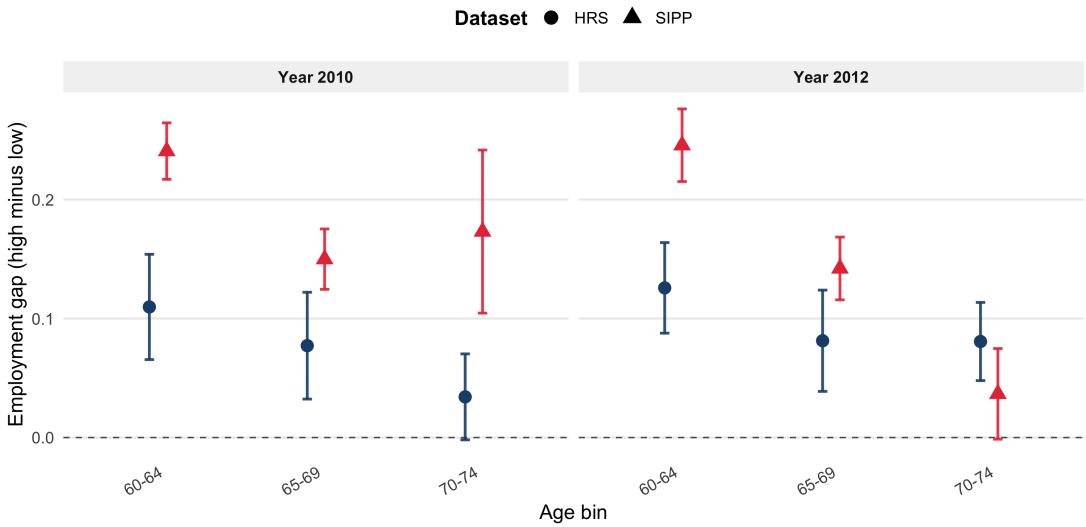
This section asks a simple question: does the data move in the way our mechanism says it should? This step is necessary before turning to the structural model. A calibrated model can fit average employment levels for many different reasons, including the wrong ones. What we need first are a few transparent reduced-form comparisons around the 2011–2012 payroll-tax holiday that tell us whether the key mechanism is present in the data. Our mechanism says that behavior should depend on the size of the tax change workers face and on how tightly payroll taxes are linked to future Social Security benefits. Empirically, we study this with a sequence of common-shock exposure designs: the holiday provides the common national shock, and predetermined worker characteristics determine how strongly that shock loads onto effective work incentives. Because the model and the later calibration are organized around the extensive margin, we focus here on retirement transitions and employment rather than on detailed hours responses.

## 4.1 Why This Exercise Matters

All estimates in this section use the monthly SIPP window from January 2010 through December 2012. The holiday period is 2011–2012, and the pre-holiday period is calendar year 2010. In each case, the treatment takes the form $\text{Post}_t \times s_i$, where $\text{Post}_t$ captures the common timing of the payroll-tax cut and $s_i$ is a predetermined exposure measure: baseline tax exposure, pre-policy claiming status, or baseline $\lambda_0$. Our main transition outcome is the monthly retirement-onset hazard: the probability that a worker who was not in the joint state of claiming benefits and working very little in month $t-1$ moves into that state in month $t$. You can think of this as a month-by-month probability of newly entering retirement. In one specification we instead use an employment indicator, which is the stock version of the same extensive-margin decision.

Before turning to the main reduced-form estimates, we document one descriptive fact that motivates the survival dimension of the model. In both the HRS and the SIPP, older workers with higher survival expectations are more likely to be employed. This does not by itself identify a causal effect, but it shows why survival is an economically meaningful source of heterogeneity in late-career labor supply.

Figure 3: Employment Gaps by Survival and Age Bin in HRS and SIPP




*Notes:* Each panel reports the high-minus-low survival coefficient from a within-year, within-age-bin regression for an employment indicator. Age bins are 60–64, 65–69, and 70–74. HRS high survival is defined by the within-cell median of subjective 10-year survival probability. SIPP high survival is defined by the within-cell median of transported baseline survival proxy ($S_i$). Error bars are 95% confidence intervals.

## 4.2 Three Reduced-Form Patterns the Model Must Explain

The rest of the section isolates three empirical patterns. Each one corresponds to a different part of the model's logic, and each one uses the same common-shock exposure idea with a different choice of $s_i$. The first is a broad reduced-form comparison based on baseline tax exposure. The second asks whether the response is larger when payroll taxes are less benefit-linked. The third asks whether, among workers who are still unclaimed, the response varies with the strength of marginal benefit accrual itself.

### Pattern 1: Workers with larger baseline exposure show larger retirement changes around the holiday

The holiday lowered the employee OASDI payroll-tax rate by 2 percentage points for everyone. But the dollar size of the cut was not the same for everyone. Workers with higher covered earnings, up to the taxable maximum, received a larger tax reduction. If payroll taxes affect retirement timing, then workers with larger baseline exposure should change behavior more.

We split workers at the median of their predetermined 2010 payroll-tax exposure and estimate a difference-in-differences regression:

$$Y_{it} = \alpha_i + \lambda_{ym} + \beta\,(\text{Post}_t \times \text{HighExposure}_i) + \Gamma_a + \varepsilon_{it}.$$

Here $Y_{it}$ is the retirement-onset hazard, $\alpha_i$ are person fixed effects, and $\lambda_{ym}$ are year-month fixed effects. The coefficient $\beta$ answers a simple question: after January 2011, did the high-exposure group change more than the low-exposure group?

Table 3 shows that the answer is yes in the data. In the baseline specification, high-exposure workers experienced a 0.73 percentage-point larger increase in the monthly retirement-onset hazard during the holiday than low-exposure workers did. Relative to a baseline monthly hazard of about 2 percent, that is a sizable shift. At the same time, we do not treat this first comparison as a standalone causal estimate. The high- and low-exposure groups differ mechanically in baseline earnings, and the pre-period is short, so this design is best read as suggestive reduced-form evidence that the holiday mattered and that the response was larger where the dollar size of the tax cut was larger. This is useful background for the rest of the section, but it is not the paper's strongest evidence.

Table 3: Baseline Tax Exposure and Retirement-Onset Hazard Around the Holiday

| | (1) Baseline | (2) + PreClaimer | (3) + Demographics | (4) Alt FE |
|---|---|---|---|---|
| *Panel A: 2011–2012 vs 2010* | | | | |
| Post × HighExposure | 0.0073*** | 0.0070*** | 0.0089*** | 0.0090*** |
| SE | (0.0024) | (0.0025) | (0.0026) | (0.0026) |
| Post × PreClaimer | | ✓ | ✓ | ✓ |
| Post × Baseline demographics (2010) | | | ✓ | ✓ |
| Age FE ($i(age)$) | ✓ | ✓ | ✓ | ✓ |
| Individual FE | ✓ | ✓ | ✓ | ✓ |
| Year-month FE | ✓ | ✓ | ✓ | |
| Year FE + Month FE | | | | ✓ |
| Observations | 77,142 | 77,142 | 77,142 | 77,142 |
| Individuals | 3,460 | 3,460 | 3,460 | 3,460 |
| *Panel B: 2013 vs 2011–2012* | | | | |
| Post × HighExposure | -0.0026 | -0.0040 | -0.0033 | -0.0035 |
| SE | (0.0036) | (0.0037) | (0.0040) | (0.0040) |
| Post × PreClaimer | | ✓ | ✓ | ✓ |
| Post × Baseline demographics (2010) | | | ✓ | ✓ |
| Age FE ($i(age)$) | ✓ | ✓ | ✓ | ✓ |
| Individual FE | ✓ | ✓ | ✓ | ✓ |
| Year-month FE | ✓ | ✓ | ✓ | |
| Year FE + Month FE | | | | ✓ |
| Observations | 61,492 | 61,492 | 61,492 | 61,492 |
| Individuals | 2,860 | 2,860 | 2,860 | 2,860 |

Notes: Outcome is the monthly retirement-onset hazard: the probability of transitioning into claim$= 1$ and monthly hours $\leq 20$, conditional on not being in that state at $t-1$. Post $= 1$ for 2011–2012 and Post $= 0$ for 2010. High exposure is the median split of baseline 2010 payroll-tax bite. Ages between 62 and 70. Standard errors are clustered by person. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

**Pattern 2: The response is larger when payroll taxes are less tied to future benefits**

A statutory payroll-tax cut is not the same as an effective tax cut for every worker. For workers who had already claimed Social Security, current payroll taxes are close to a pure tax: paying the tax does not materially raise future benefits. For workers who had not yet claimed, an extra year of covered earnings can still raise future benefits, so part of the payroll tax is offset by benefit accrual. If benefit linkage matters, pre-claimers should respond more strongly to the same statutory tax change.

We compare pre-claimers and non-claimers before and after the holiday:

$$\begin{aligned} Y_{it} = \alpha_i + \lambda_{ym} &+ \beta_1 \left(\text{Post}_t \times \text{PreClaimer}_i\right) \\ &+ \beta_2 \left(\text{Post}_t \times \text{Survival}_i^c\right) \\ &+ \beta_3 \left(\text{Post}_t \times \text{PreClaimer}_i \times \text{Survival}_i^c\right) + \varepsilon_{it}. \end{aligned}$$

Here $Y_{it}$ is our main employment indicator, $\text{PreClaimer}_i$ equals one for workers already claiming benefits before the holiday, and $\text{Survival}_i^c$ is baseline survival centered to have mean zero. The main coefficient, $\beta_1$, is another difference-in-differences estimate. It asks whether pre-claimers changed employment more than non-claimers did after the holiday, after netting out permanent differences across people and common shocks over time. Because survival is centered, $\beta_1$ is the pre-claimer effect for a worker with average survival.

Table 4 shows that pre-claimers did respond more strongly. In the employment column, pre-claimers' employment changed by 7.6 percentage points more than the employment of comparable non-claimers between 2010 and 2011–2012. The hours column moves in the same direction, but the main lesson is the extensive-margin one: the response is larger when payroll taxes are less benefit-linked. This second pattern is central for the paper because it is more closely tied to the mechanism than the broad exposure comparison in Pattern 1.

Table 4: Claiming-Status Difference-in-Differences

| | Employment indicator | Monthly hours |
|---|---|---|
| Post × Pre-claimer ($\hat{\beta}$) | 0.076*** | 13.69*** |
| | [0.064, 0.089] | [11.69, 15.70] |
| Survival interactions included | ✓ | ✓ |
| Person & year-month FE | ✓ | ✓ |
| Person-month observations | 236,162 | 236,162 |
| Unique individuals | 8,499 | 8,499 |

Notes: Sample is the cut episode (2010 vs. 2011–2012). The specification also includes Post× centered survival probability and Post× Pre-claimer × centered survival probability, although only Post× Pre-claimer is displayed. Because survival is centered, the reported coefficient is the pre-claimer effect at average survival. Employment is the main extensive-margin outcome; hours are shown as a secondary corroborating measure. Ages between 62 and 70. Brackets report 95% confidence intervals clustered by person. *** $p < 0.01$.

**Pattern 3: Among unclaimed workers, the response varies with marginal benefit accrual**

Non-claimers are not all the same. Some workers get much more future Social Security benefit accrual from one more year of earnings than others do. We summarize that mechanical accrual incentive with $\lambda_0$. Formally, $\lambda_0$ is the marginal increase in future annual Social Security benefits generated by one more year of covered earnings. A higher $\lambda_0$ means that an extra dollar of payroll tax is more strongly offset by future benefits, so the worker starts from a smaller effective tax wedge. When the statutory payroll tax falls by the same 2 percentage points for everyone, the proportional change in that effective wedge is larger for high-$\lambda_0$ workers. If the model is correct, their retirement timing should shift more.

This is why we move from a binary comparison to a continuous one. The earlier claiming-status result asks whether workers on two different sides of the benefit-linkage margin respond differently. This specification goes one step deeper. It asks whether, within the unclaimed group itself, the holiday response changes smoothly with the strength of benefit accrual. That is a sharper test of the model because $\lambda_0$ is the actual object that appears in the mechanism, not just a proxy group indicator.

We therefore restrict attention to unclaimed workers with observed earnings and replace

the binary group indicator with the continuous accrual measure:

$$Y_{it} = \alpha_i + \lambda_{ym} + \beta \left(\text{Post}_t \times \lambda_{0,i}^c\right) + \text{Post}_t \times X_i'\gamma + \Gamma_a + \varepsilon_{it}.$$

Here $Y_{it}$ is the monthly retirement-onset hazard for worker $i$ in month $t$, estimated only on person-months in which the worker is still at risk of newly entering retirement. $\alpha_i$ are person fixed effects, which absorb permanent differences across workers. $\lambda_{ym}$ are year-month fixed effects, which absorb shocks common to everyone in a given month, including the overall timing of the holiday. $\Gamma_a$ are age fixed effects. $\lambda_{0,i}^c$ is centered marginal benefit accrual, so it measures how far worker $i$'s accrual rate is above or below the sample average. $X_i$ includes centered survival and, in richer specifications, baseline demographics. Because those controls are fixed over time, they are absorbed by the person fixed effects in levels; what we can identify is whether workers with different baseline characteristics changed differently after the policy, which is why they enter interacted with $\text{Post}_t$.

The coefficient $\beta$ is the key object. Econometrically, this is a continuous-treatment difference-in-differences. Instead of asking whether a "treated group" changed more than a "control group," it asks whether the holiday effect gets larger as $\lambda_0$ rises. If two otherwise similar unclaimed workers differ in $\lambda_0$ by 0.01, then the model predicts that their holiday response should differ by approximately $0.01 \times \beta$. Since the outcome is a hazard indicator, $\beta$ is measured in probability units.

Table 5 shows a clear positive relationship. In the baseline specification, $\hat{\beta} = 1.0647$ (SE $0.3601$, $p = 0.003$). The estimate remains positive and statistically significant when baseline demographics are added and when year-month fixed effects are replaced with separate year and month fixed effects. Using the interquartile range of $\lambda_0$ in the sample, about 0.007, the baseline estimate implies about a $1.0647 \times 0.007 \approx 0.75$ percentage-point larger increase in the monthly retirement-onset hazard for a worker at the 75th percentile of $\lambda_0$ than for a worker at the 25th percentile. Relative to the baseline monthly retirement-onset hazard in this sample, that is economically large.

The sign deserves interpretation. A positive $\hat{\beta}$ does not mean that high-$\lambda_0$ workers face a larger tax in levels. It means that the holiday produced a larger proportional change in their effective tax wedge. To see this, write the effective wedge as

$$\tau_i^{\text{eff}} = \tau^{\text{OASDI}} - \lambda_{0,i}.$$

Consider two workers of the same age. If one worker has $\lambda_0 = 0.022$, her effective wedge falls

Table 5: Benefit Accrual ($\lambda_0$) and Retirement-Onset Hazard Among Unclaimed Workers

| | (1) Baseline | (2) + Demographics | (3) Alt FE |
|---|---|---|---|
| Post $\times$ $\lambda_0$ (centered) | 1.0647*** | 0.7572** | 0.7519** |
| SE | (0.3601) | (0.3742) | (0.3733) |
| 95% CI | [0.3589, 1.7705] | [0.0238, 1.4906] | [0.0202, 1.4835] |
| Post $\times$ Survival (centered) | ✓ | ✓ | ✓ |
| Post $\times$ Demographics (2010) | | ✓ | ✓ |
| Age FE | ✓ | ✓ | ✓ |
| Individual FE | ✓ | ✓ | ✓ |
| Year-month FE | ✓ | ✓ | |
| Year FE + Month FE | | | ✓ |
| Observations | 44,321 | 44,321 | 44,321 |
| Individuals | 1,758 | 1,758 | 1,758 |

Notes: Sample is unclaimed workers aged 62–70 with observed (non-imputed) earnings. Outcome is the monthly retirement-onset hazard, estimated on person-months at risk of newly entering retirement. Columns report weighted linear probability models. $\lambda_0$ is the mechanical derivative of annual Social Security benefits with respect to annual covered earnings, evaluated at baseline 2010 AIME and accounting for top-35 selection, bend points, and age-specific actuarial adjustments. Standard errors are clustered by person. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

from $0.062 - 0.022 = 0.040$ before the holiday to $0.042 - 0.022 = 0.020$ during the holiday, a 50% reduction. If another worker has $\lambda_0 = 0.004$, her effective wedge falls from $0.058$ to $0.038$, only a 34% reduction. The same statutory reform therefore produces a larger effective-price change for the high-$\lambda_0$ worker, which is exactly what the positive estimate is capturing.

This is why this third pattern matters for the paper. It is not just another heterogeneity result. It is the reduced-form counterpart to the model's central state variable. If the model is right, it should be able to explain not only that behavior moved around the holiday, and not only that claimers and non-claimers differ, but also that the response varies continuously with the strength of the benefit-accrual channel itself.

### 4.3 Why These Empirical Facts Matter

These reduced-form estimates are necessary because they tell us whether the mechanism is in the data before we ask the model to quantify it. We will discipline the model by pre-reform employment moments, not by the holiday coefficients themselves. That is exactly why the reduced-form exercise is important. Without it, the model could fit 2010 employment levels for the wrong reason. The three patterns above provide empirical benchmarks that any credible version of the model should be able to explain: workers with larger baseline exposure

moved more around the holiday, the response is stronger when payroll taxes are less benefit-linked, and the response varies systematically with the strength of marginal benefit accrual. The first pattern is broad and suggestive. The second and third are more tightly connected to the mechanism and therefore carry more weight in the paper's interpretation.

Additional timing diagnostics and robustness results are reported in Section B. The appendix reports a coefficient event study for the tax-exposure design and companion group-path figures for the claiming-status and $\lambda_0$ designs, along with the broader placebo and robustness exercises. Those results are useful, but they are not the core of the paper. The core is the pattern across the three comparisons, especially the latter two, because they map most directly into the extensive-margin mechanism that the model is built to capture.

# 5 A Dynamic Retirement Model of Seniors

Our data track employment among workers in their 60s who are already near retirement. We therefore do not need a full lifecycle model from youth to death. Instead, we build a dynamic retirement model that begins at age 62 and takes earlier earnings and wealth as given. The model focuses on the decisions that matter at this stage of life: whether to keep working, when to stop, and how delaying benefit receipt changes the calculus. Retirement is irreversible—once a worker exits, she does not return.

Why start at 62? By that age, most households have already built up the savings that will shape their retirement timing. Over the short horizon we study, the main role of assets is to capture differences in starting wealth, not to be re-optimized year by year. Workers enter the model at age 62 with a fixed asset type; we suppress the asset index in what follows and reintroduce it when we parameterize the model.

## 5.1 Extensive-Margin Dynamics: Effective Wedge and Option Value

What is the payoff to working one more period? That payoff differs across workers because of Social Security. A worker who has not yet claimed benefits earns two things by staying on the job—current after-tax wages and higher future benefits. A worker who has already claimed gets only the after-tax wage; worse, if she is below Full Retirement Age (FRA), the Retirement Earnings Test (RET) can reduce even that. We now formalize these forces.

### 5.1.1 Continuation payoff and normalized wedge

Consider a worker of age $a_0$ with mortality type $\pi$ and claiming status $g \in \{c, u\}$ (claimed or unclaimed). Time is discrete: $t \in \{0, 1, \ldots, N\}$, where $t = N$ is mandatory exit. Each period, the worker decides whether to continue working or to retire permanently. She survives from $t$ to $t+1$ with probability $\sigma_t(\pi) \in (0, 1)$. As before, the annuity factor at date $t$ is the present value of \$1 per period for the rest of life:

$$\mu_t(\pi) \equiv \sum_{j=1}^{N-t} \beta^j \prod_{k=0}^{j-1} \sigma_{t+k}(\pi), \tag{13}$$

with discount factor $\beta \in (0, 1)$. Workers who expect to live longer (higher $\pi$) have a larger $\mu_t$, because more future payments are likely to be collected.

**The continuation payoff.** The dollar-valued payoff from one more period of work, beyond the after-tax wage itself, is

$$A_g(t, e, \pi) \equiv \underbrace{w(e)\lambda_g(e)\mu_t(\pi)}_{\text{benefit accrual value}} - \underbrace{w(e)\theta\, \mathbf{1}\{g = c,\ a_t < \text{FRA},\ w(e) > \bar{y}\}}_{\text{RET penalty for claimers before FRA}}. \tag{14}$$

The first term captures the future benefits earned by working today. For unclaimed workers, $\lambda_u(e) = \lambda_0(e) > 0$—each extra dollar of earnings raises the annual benefit by $\lambda_0$, and those higher benefits are collected for an expected $\mu_t$ periods. For claimed workers, $\lambda_c(e) = 0$: they have already locked in their benefit, so additional work does not raise it. The second term captures the RET: a claimed worker below FRA whose earnings exceed the exempt amount $\bar{y}$ faces a proportional penalty $\theta \geq 0$ on current earnings.

Dividing by the wage gives a normalized wedge:

$$\kappa_g(t, e, \pi) \equiv \frac{A_g(t, e, \pi)}{w(e)} = \lambda_g(e)\mu_t(\pi) - \theta\, \mathbf{1}\{g = c,\ a_t < \text{FRA},\ w(e) > \bar{y}\}. \tag{15}$$

The effective payroll-tax wedge is then

$$\tau_g^{\text{eff}}(t, e, \pi) \equiv \tau - \kappa_g(t, e, \pi). \tag{16}$$

This is the key object in the paper. Everyone faces the same statutory tax rate $\tau$, but the effective tax on work differs. For an unclaimed worker with strong benefit accrual, $\kappa_u$ is large, so $\tau_u^{\text{eff}}$ can fall well below the statutory rate—and may even turn negative if the accrual value

exceeds $\tau$. For a claimed worker below FRA, the RET adds to the tax burden, pushing $\tau_c^{\text{eff}}$ above the statutory rate.

The difference in normalized wedges across claiming status is

$$\kappa_u(t, e, \pi) - \kappa_c(t, e, \pi) = \lambda_0(e)\mu_t(\pi) + \theta\, \mathbf{1}\{a_t < \text{FRA},\ w(e) > \bar{y}\}. \tag{17}$$

Before FRA, the gap has two parts: unclaimed workers gain from benefit accrual while claimed workers lose from the RET. After FRA, the RET no longer applies and the gap reduces to pure accrual: $\lambda_0(e)\mu_t(\pi)$.

**From individuals to the population.** To see how much dispersion Social Security creates in aggregate, let $m_t(s)$ be the mass of active workers at age $t$ in state $s$ (which collects claiming status, earnings type, survival type, asset type, and labor-market shock). The cross-sectional distribution of effective wedges at age $t$ is

$$F_t^{\tau}(x) \equiv \frac{\sum_s m_t(s)\, \mathbf{1}\{\tau^{\text{eff}}(s) \le x\}}{\sum_s m_t(s)}. \tag{18}$$

In the calibration, we plot this distribution for workers ages 62–70 under the common 6.2 percent statutory rate. Even though the statutory rate is a single number, claiming rules, benefit accrual, and the RET spread the effective wedge across a wide range.

**Flow payoff from continued work.** Putting together after-tax wages and the continuation payoff, the total period-$t$ return to working is

$$\Pi_g(t, \nu_t, \pi, e) \equiv w(e)(1-\tau) + A_g(t, e, \pi) - \nu_t = w(e)\big(1 - \tau + \kappa_g(t, e, \pi)\big) - \nu_t, \tag{19}$$

where $\nu_t \ge 0$ is the disutility of work in that period. The expression $1 - \tau + \kappa_g$ is the effective share of each dollar of earnings that the worker keeps (including the value of future benefits), so $\Pi_g$ is simply the effective wage minus the cost of showing up. For notational convenience, we suppress the earnings-type argument $e$ going forward and write $w$, $\lambda_0$, $A_g(t, \pi)$, and $\kappa_g(t, \pi)$.

**Assumption 1.** *Claimed and unclaimed workers are comparable in observables and face the same non-accrual retirement environment. Conditional on earnings type, survival type, and predetermined asset type, the primitive payoff and transition objects are common across groups except for the continuation term $A_g(t, \pi)$, equivalently the normalized wedge $\kappa_g(t, \pi)$. Differences in the cross-*

*sectional distribution of latent work costs at date $t$ may still arise endogenously through past selection into continued employment.*

**Irreversible Retirement.** Once a worker exits at date $t$, the worker remains non-employed for all $s > t$. Let $V^R(t)$ denote the common non-accrual value of exit. Then

$$V_c^R(t) = V_u^R(t) = V^R(t). \tag{20}$$

This normalization does not assert that claimed and unclaimed retirees have identical total Social Security wealth. Instead, for the purpose of the one-more-period work decision, benefit-level differences that are not created by current work are treated as already netted out. The common retirement value is therefore a direct consequence of the accounting convention in the flow payoff (19).

Because the present value of the marginal benefit accrual from one additional period of work is already embedded in $\Pi_g$ through the term $A_g$, the retirement value needs to capture only the non-accrual components of post-retirement: leisure value, predetermined asset position, non–Social Security pension wealth, spousal income, and any other factors that are common across claiming groups within a given asset type under Assumption 1. There is no residual accrual value that differs across $g$ at the moment of exit, because that value has been accounted for period-by-period in the flow payoff. This normalization avoids double counting: the marginal benefit of current work appears exactly once, in $\Pi_g$, and not again in $V^R$. Section 5.5 returns to how this benchmark relates to a richer model with endogenous claiming.

## 5.2 Continuation thresholds and monotone comparative statics

Let $V_g^W(t, \nu_t, \pi; \tau)$ denote the value of an employed worker with claiming status $g$ at date $t$. For $t \in \{0, \ldots, N-1\}$,

$$V_g^W(t, \nu_t, \pi; \tau) = \max\left\{ \underbrace{V^R(t)}_{\text{exit now}}, \quad \underbrace{\Pi_g(t, \nu_t, \pi)}_{\text{current flow payoff}} + \underbrace{\beta \sigma_t(\pi)\, \mathbb{E}_t\big[V_g^W(t{+}1, \nu_{t+1}, \pi; \tau)\big]}_{\text{discounted continuation value}} \right\}, \tag{21}$$

with terminal condition

$$V_g^W(N, \nu_N, \pi; \tau) = V^R(N).$$

The option value of continued work is

$$\mathcal{O}_g(t, \nu_t, \pi; \tau) \equiv \Pi_g(t, \nu_t, \pi) + \beta\sigma_t(\pi)\,\mathbb{E}_t\left[V_g^W(t+1, \nu_{t+1}, \pi; \tau)\right] - V^R(t).$$

The worker continues if and only if $\mathcal{O}_g(t, \nu_t, \pi; \tau) \geq 0$. Because the current work-cost state enters additively, the continuation rule is threshold based:

$$\nu_t \leq \nu_g^*(t, \tau, \pi) \equiv w(1-\tau) + A_g(t, \pi) + \Delta_g(t, \tau, \pi), \tag{22}$$

where

$$\Delta_g(t, \tau, \pi) \equiv \beta\sigma_t(\pi)\,\mathbb{E}_t\left[V_g^W(t+1, \nu_{t+1}, \pi; \tau)\right] - V^R(t) \tag{23}$$

is the dynamic continuation premium.

**Proposition 1.** *Under Assumption 1, and with $\sigma_t(\pi)$ weakly increasing in $\pi$ for every $t$:*

1. *For every $t \in \{0, \ldots, N-1\}$ and every $\pi$,*

$$\nu_u^*(t, \tau, \pi) > \nu_c^*(t, \tau, \pi). \tag{24}$$

   *Hence unclaimed workers tolerate strictly higher work disutility before exiting.*

2. *For every $\pi' \geq \pi$ and each group $g \in \{c, u\}$,*

$$\nu_g^*(t, \tau, \pi') \geq \nu_g^*(t, \tau, \pi), \tag{25}$$

   *and the threshold gap is weakly increasing in survival type:*

$$\left[\nu_u^*(t, \tau, \pi') - \nu_c^*(t, \tau, \pi')\right] \geq \left[\nu_u^*(t, \tau, \pi) - \nu_c^*(t, \tau, \pi)\right]. \tag{26}$$

3. *For every $\tau' \leq \tau$ and each $g \in \{c, u\}$,*

$$\nu_g^*(t, \tau', \pi) \geq \nu_g^*(t, \tau, \pi). \tag{27}$$

   *Thus a payroll tax cut weakly raises the continuation threshold, making workers more willing to stay employed.*

*Proof.* See Appendix A. □

Equation (22) shows that survival affects the continuation threshold through two objects:

the direct accrual term $A_g(t,\pi)$ and the dynamic continuation premium $\Delta_g(t,\tau,\pi)$. The direct accrual component contributes $w\lambda_0(e)\mu_t'(\pi)$ only for unclaimed workers. Thus, holding the continuation-premium response fixed, higher survival raises the unclaimed continuation threshold more strongly than the claimed continuation threshold. We use this as the economic intuition behind the survival interaction below rather than as a separate theorem under the current assumptions. Whenever $\Delta_g(t,\tau,\pi)$ is differentiable in $\tau$, the threshold sensitivity is

$$\frac{\partial \nu_g^*}{\partial \tau} = -w + \frac{\partial \Delta_g}{\partial \tau} \leq 0. \tag{28}$$

Because $\partial \nu_g^*/\partial \tau \leq 0$, a tax cut $d\tau < 0$ raises the continuation threshold by $|d\tau| \cdot w$ through the direct flow effect, plus an additional upward shift through the continuation premium.

### 5.3 Two-Channels Decomposition

Let $F_g(\cdot \mid t)$ denote the cross-sectional distribution of the current latent work-cost state among workers who are alive and employed at the start of period $t$, with density $f_g(\cdot \mid t)$. This distribution may differ across groups because after-retiremenet decisions can select different workers into the surviving employed stock, even when the primitive latent work-cost process is common across groups conditional on $(e,\pi)$. Let $C_t = 1$ denote the event that a worker continues working from the start of period $t$ into period $t+1$.

Holding fixed the start-of-period survivor distribution $F_g(\cdot \mid t)$, the period-$t$ continuation probability is

$$P_g(C_t = 1 \mid \tau, \pi) = F_g\big(\nu_g^*(t,\tau,\pi) \mid t\big)\,. \tag{29}$$

Differentiating (29), we get

$$\frac{\partial P_g}{\partial \tau} = f_g\big(\nu_g^*(t,\tau,\pi) \mid t\big) \cdot \frac{\partial \nu_g^*}{\partial \tau}. \tag{30}$$

**Proposition 2** (Two-channel decomposition of the extensive-margin differential)**.** *The differential response of continuation probabilities to a tax change is*

$$\frac{\partial P_c}{\partial \tau} - \frac{\partial P_u}{\partial \tau} = \underbrace{w\Big[f_u(\nu_u^*) - f_c(\nu_c^*)\Big]}_{\textit{density channel}} + \underbrace{f_c(\nu_c^*)\frac{\partial \Delta_c}{\partial \tau} - f_u(\nu_u^*)\frac{\partial \Delta_u}{\partial \tau}}_{\textit{continuation-value channel}}, \tag{31}$$

*where $(t,\tau,\pi)$ arguments are suppressed.*

*Proof.* Substitute (28) into (30) and rearrange. □

The density channel captures where claimed and unclaimed workers sit on the distribu-

tion of latent work costs at the moment of decision: a tax cut shifts both thresholds rightward, and the group whose threshold sits at a higher density gains more marginal continuers. The continuation-value channel captures how much the tax cut raises the option value of preserving future flexibility for each group.

**Lemma 1.** *For every $g$, $t$, and $\pi$,*

$$0 \leq -\frac{\partial \Delta_g(t, \tau, \pi)}{\partial \tau} \leq w \sum_{j=1}^{N-t} \beta^j \prod_{k=0}^{j-1} \sigma_{t+k}(\pi). \tag{32}$$

*Hence a tax cut weakly raises the continuation premium (the option value of staying employed), and this gain is bounded above by the discounted, survival-weighted benefit of saving one wage unit of tax in each remaining period.*

*Proof.* The lower bound follows from Proposition 1(iii): a lower tax weakly raises all future continuation values, so $\partial \Delta_g / \partial \tau \leq 0$. For the upper bound, compare the value effect of a marginal tax cut to the counterfactual path in which the worker continues working in every future period. The tax cut can raise the flow payoff by at most $w$ in any future period, and future gains are discounted and survival weighted, which gives (32). □

For workers near retirement, the horizon $N - t$ is short, so Lemma 1 implies that the continuation-value increase from a tax cut is quantitatively small relative to the direct one-period flow effect that underlies the density channel. This formalizes the idea that, in older samples such as workers aged 62 and above, the density channel dominates the option-value channel: the payroll tax holiday's main effect on retirement operates through the immediate wage gain, not through enhanced future flexibility.

Suppose first that $F_c(\cdot \mid t) = F_u(\cdot \mid t) \equiv F(\cdot \mid t)$ with density $f(\cdot \mid t)$, and that $f(\cdot \mid t)$ is unimodal with mode $\bar{\nu}(t)$. The sign of the density channel depends on where the thresholds sit relative to the mode. When both thresholds lie above the mode (on the downward-sloping side), the higher unclaimed threshold sits at a less dense point: $f(\nu_u^* \mid t) \leq f(\nu_c^* \mid t)$. But for older workers aged 62 and above, the empirically relevant case is the reverse. Among workers who have survived to these ages and are still employed, the bulk of the population has relatively low work costs—they selected into continued employment precisely because working was not too costly for them. The mode of the survivor density therefore sits well below both continuation thresholds, placing both $\nu_c^*$ and $\nu_u^*$ on the declining right tail of the density. Because $\nu_c^* < \nu_u^*$ (Proposition 1(i)) and the density is decreasing to the right of the

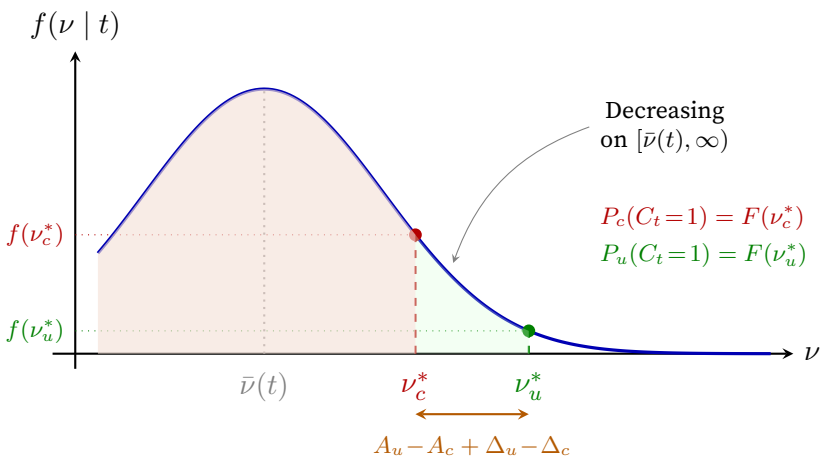



Figure 4: Continuation thresholds and the density condition for older workers. The survivor density $f(\nu \mid t)$ among workers still employed at ages 62+ peaks at a low mode $\bar{\nu}(t)$, reflecting selection: most remaining workers have low work costs. Both thresholds lie above the mode on the declining right tail. Because $\nu_c^* < \nu_u^*$ (Proposition 1(i)) and the density is decreasing, $f(\nu_c^*) > f(\nu_u^*)$: more claimed workers sit near the exit margin. A tax cut $d\tau < 0$ shifts both thresholds rightward by $w\,|d\tau|$, and the density channel implies that claimed workers gain more marginal continuers, producing a larger decline in their exit hazard. Shaded areas represent continuation probabilities conditional on employment at the start of period $t$.

mode, the claimed threshold sits at a higher density:

$$f(\nu_c^* \mid t) > f(\nu_u^* \mid t). \tag{33}$$

Intuitively, claimed workers—who lack the benefit-accrual cushion—have a lower tolerance for work disutility, placing their exit margin closer to where the remaining employed workers actually sit in the distribution. More claimed workers are therefore marginal: a common threshold shift from the tax cut moves more of them across the exit boundary.

The survival interaction reinforces this pattern. Higher survival raises the annuity factor $\mu_t(\pi)$, which directly shifts the unclaimed threshold rightward through the term $w\lambda_0\mu_t(\pi)$. Because the claimed threshold lacks this direct accrual channel, higher survival pushes $\nu_u^*$ further into the thin right tail of the density, widening the gap $f(\nu_c^*) - f(\nu_u^*)$ and strengthening the predicted differential response.

### 5.3.1 Exit hazard and the empirical prediction

Define the period-$t$ exit hazard, conditional on being employed and alive at the start of the period, as

$$h_g(t \mid \tau, \pi) \;\equiv\; 1 - F_g\big(\nu_g^*(t, \tau, \pi) \;\big|\; t\big) \;=\; 1 - P_g(C_t = 1 \mid \tau, \pi). \tag{34}$$

Since a tax cut raises the continuation threshold (Proposition 1(iii)) and therefore increases the continuation probability, the exit hazard is weakly decreasing in the size of the cut for both groups. During the payroll tax holiday, both claimed and unclaimed workers face lower exit hazards than under the baseline tax rate.

**Proposition 3.** *A tax cut $d\tau < 0$ lowers both groups' exit hazards, but claimed workers' exit hazard falls more than unclaimed workers' whenever the density channel is sufficiently strong relative to the continuation-value channel.*

*Formally, let*

$$B_t(\pi) \equiv w \sum_{j=1}^{N-t} \beta^j \prod_{k=0}^{j-1} \sigma_{t+k}(\pi)$$

*denote the maximum possible sensitivity of the continuation premium to the tax (Lemma 1). This quantity captures the largest gain in the option value of staying employed that a tax cut can deliver: it equals the discounted, survival-weighted benefit of saving one wage unit of tax in every remaining period. If the density gap between claimed and unclaimed workers exceeds this maximum option-value effect,*

$$\underbrace{w\Big[f_c(\nu_c^* \mid t) - f_u(\nu_u^* \mid t)\Big]}_{\textit{density-channel advantage}} \;>\; \underbrace{f_u(\nu_u^* \mid t)\, B_t(\pi)}_{\textit{max. continuation-value distortion}}, \tag{35}$$

*then for any $d\tau < 0$,*

$$\frac{\partial h_c}{\partial \tau} \;>\; \frac{\partial h_u}{\partial \tau} \;\geq\; 0 \qquad \Longrightarrow \qquad dh_c < dh_u \leq 0. \tag{36}$$

*That is, both groups' exit hazards decline during the tax holiday, but claimed workers—who lack the benefit-accrual cushion and therefore sit closer to the margin—experience a strictly larger decline. Because $B_t(\pi)$ shrinks as the remaining horizon $N - t$ shortens, the sufficient condition is easy to satisfy for workers near retirement, where there is little future option value left for the tax cut to affect.*

*Proof.* By definition, $\partial h_g/\partial \tau = -\partial P_g/\partial \tau$. From Lemma 1, $\partial \Delta_u/\partial \tau \geq -B_t(\pi)$ and $-\partial \Delta_c/\partial \tau \geq$

0. Proposition 2 gives

$$\begin{aligned}\frac{\partial P_c}{\partial \tau} - \frac{\partial P_u}{\partial \tau} &\leq w\Big[f_u(\nu_u^*) - f_c(\nu_c^*)\Big] - f_u(\nu_u^*)\frac{\partial \Delta_u}{\partial \tau} \\ &\leq w\Big[f_u(\nu_u^*) - f_c(\nu_c^*)\Big] + f_u(\nu_u^*)\, B_t(\pi).\end{aligned}$$

Under the density condition $f_c(\nu_c^*) > f_u(\nu_u^*)$ and condition (35) makes the right-hand side strictly negative, so $\partial P_c/\partial\tau < \partial P_u/\partial\tau$. Hence $\partial h_c/\partial\tau > \partial h_u/\partial\tau$. Multiplying by $d\tau < 0$ reverses the inequality: $dh_c < dh_u$. Non-negativity of the derivatives follows from Proposition 1(iii), so $dh_g \leq 0$. □

This proposition translates the threshold model directly into the payroll tax holiday setting: during the 2011–2012 tax cut, both groups' exit hazards fall, but claimed workers—who face a pure wage tax with no offsetting benefit accrual and whose exit threshold sits at a denser point in the survivor distribution—see a strictly larger decline in their exit hazard.

### 5.4 Employment stock dynamics and policy-sequence interpretation

A defining feature of the senior labor market is that retirement is largely irreversible: workers who exit employment rarely return. This asymmetry between entry and exit distinguishes the retirement margin from labor supply adjustments earlier in the life cycle, where workers routinely move in and out of employment. Our threshold model embeds this irreversibility through absorbing exit, and it carries a sharp implication: a temporary policy shock can have persistent effects on the employment stock, even after the policy reverts. This subsection develops that implication formally. The result is useful not only for interpreting the specific time path of employment around the 2011–2012 payroll tax holiday—in particular, why the end of the holiday need not produce a symmetric reversal—but also for understanding more generally how irreversibility shapes the propagation of fiscal incentives in aging labor markets.

Let $\ell_g(t \mid \tau, \pi)$ denote the probability that a worker of type $(g, \pi)$ is alive and employed at the start of period $t$, conditional on employment at $t = 0$. With absorbing exit and no re-entry,

$$\ell_g(t+1 \mid \tau, \pi) = \ell_g(t \mid \tau, \pi) \cdot P_g(C_t = 1 \mid \tau, \pi). \tag{37}$$

**Proposition 4** (Tax holiday generates propagating stock gaps)**.** *Fix $(g, \pi)$ and consider a temporary tax cut at date $t$: the baseline path keeps the tax rate at $\tau$ in every period, while the holiday path sets $\tau_t = \tau - \delta$ with $\delta > 0$ at date $t$ and reverts to $\tau$ thereafter.*

1. *(Impact.) The tax cut raises the continuation threshold, so fewer workers exit at date $t$. The holiday path therefore has strictly higher employment at $t{+}1$ than the baseline.*

2. *(Propagation.) Because exit is irreversible, the extra workers retained at date $t$ remain employed in all subsequent periods (until they exit for other reasons). The employment gap evolves as*

$$\ell_g^{\text{hol}}(t{+}s \mid \pi) - \ell_g^{\text{base}}(t{+}s \mid \pi) = \Big[\ell_g^{\text{hol}}(t{+}1 \mid \pi) - \ell_g^{\text{base}}(t{+}1 \mid \pi)\Big] \prod_{r=1}^{s-1} P_g(C_{t+r} = 1 \mid \tau, \pi) : \quad (38)$$

*the initial gain is carried forward and shrinks only through subsequent exit, never through reversal.*

3. *(Reversion.) Once the tax reverts to $\tau$, continuation probabilities are the same on both paths, so the stock gap stops growing but does not disappear. If the tax overshoots to $\tau{+}\epsilon$ upon restoration, a negative shock hits a workforce that has already been selected by the holiday.*

*Proof.* We prove each part in turn.

*Part (i):* By Proposition 1(iii), the tax cut raises the continuation threshold: $\nu_g^*(t, \tau - \delta, \pi) \geq \nu_g^*(t, \tau, \pi)$. Since a worker continues if and only if $\nu_t \leq \nu_g^*$, the continuation probability satisfies

$$P_g(C_t = 1 \mid \tau - \delta, \pi) \;=\; F_g\big(\nu_g^*(t, \tau - \delta, \pi) \mid t\big) \;\geq\; F_g\big(\nu_g^*(t, \tau, \pi) \mid t\big) \;=\; P_g(C_t = 1 \mid \tau, \pi),$$

with strict inequality whenever $f_g > 0$ at the threshold. The two paths coincide up to date $t$, so $\ell_g^{\text{hol}}(t \mid \pi) = \ell_g^{\text{base}}(t \mid \pi)$. Applying the stock recursion (37),

$$\begin{aligned} \ell_g^{\text{hol}}(t{+}1 \mid \pi) &= \ell_g^{\text{hol}}(t \mid \pi) \;\cdot\; P_g(C_t{=}1 \mid \tau{-}\delta, \pi) \\ &> \ell_g^{\text{base}}(t \mid \pi) \;\cdot\; P_g(C_t{=}1 \mid \tau, \pi) \\ &= \ell_g^{\text{base}}(t{+}1 \mid \pi). \end{aligned} \quad (39)$$

*Part (ii):* After date $t$ the tax reverts to $\tau$ on both paths, so the continuation probabilities coincide for every $r \geq 1$. Write $\mathcal{P}_s \equiv \prod_{r=1}^{s-1} P_g(C_{t+r} = 1 \mid \tau, \pi)$ for the common post-holiday survival factor. Iterating (37) from $t{+}1$ to $t{+}s$ on each path,

$$\begin{aligned} \ell_g^{\text{hol}}(t{+}s \mid \pi) &= \ell_g^{\text{hol}}(t{+}1 \mid \pi)\, \mathcal{P}_s, \\ \ell_g^{\text{base}}(t{+}s \mid \pi) &= \ell_g^{\text{base}}(t{+}1 \mid \pi)\, \mathcal{P}_s. \end{aligned}$$

Subtracting,

$$\ell_g^{\text{hol}}(t{+}s \mid \pi) - \ell_g^{\text{base}}(t{+}s \mid \pi) \;=\; \underbrace{\left[\ell_g^{\text{hol}}(t{+}1 \mid \pi) - \ell_g^{\text{base}}(t{+}1 \mid \pi)\right]}_{>0 \text{ by } (39)} \cdot \mathcal{P}_s, \tag{40}$$

which is (38). Each factor in $\mathcal{P}_s$ lies in $(0, 1]$, so the gap is strictly positive for all $s \geq 1$ but weakly shrinking. Crucially, the gap decays only through subsequent exit—never through reversal—because workers retained by the holiday cannot be selectively removed once the tax reverts.

*Part (iii):* When the tax reverts to $\tau$ at date $t{+}1$, the continuation probabilities are identical on both paths, so no new gap accrues. The stock gap from (40) persists at level $[\ell_g^{\text{hol}}(t{+}1 \mid \pi) - \ell_g^{\text{base}}(t{+}1 \mid \pi)] \cdot \mathcal{P}_s$, shrinking only through attrition.

If instead the tax is raised to $\tau + \epsilon$ upon restoration, the continuation probability at $t{+}1$ falls below its baseline level $P_g(C_{t+1} = 1 \mid \tau, \pi)$. This negative shock acts on the holiday-path stock $\ell_g^{\text{hol}}(t{+}1 \mid \pi)$, which is both larger and—by virtue of tax holiday selection—composed of workers with systematically lower work costs than those on the baseline path. □

This proposition maps directly onto the 2011–2012 payroll tax holiday. During the holiday, the 2 percentage point tax cut raises continuation probabilities and builds an employment stock gap relative to the no-holiday counterfactual. When the tax reverts at the start of 2013, the accumulated gap persists but no longer grows. The historical episode can display asymmetric employment changes across the holiday and restoration phases because the restoration acts on a stock that is two years later and whose survivor distribution has already been shaped by the tax holiday selection.

### 5.5 Discussion

**Option values versus reservation wages.** The distinction between option values and reservation wages clarifies why survival heterogeneity matters for welfare in a way that a static framework cannot capture. In a standard static model, a worker exits when the net wage falls below a reservation wage $w^R$; the welfare cost of a tax-induced exit is proportional to the flow surplus $w(1 - \tau) - w^R$, which is approximately zero for the marginal worker. Mortality heterogeneity plays no role in this calculation because the reservation wage reflects only the current-period comparison of work and leisure values. In the dynamic model with irreversible exit, the relevant object is the option value $\mathcal{O}_g(t, \nu_t, \pi)$, which embeds both the value of preserving future flexibility—the chance that a favorable disutility draw arrives in later

periods—and, for unclaimed workers, the survival-weighted benefit accrual $\lambda_0 \mu_t(\pi)$ that accumulates over the remaining work horizon. A worker near the exit margin may have a small current flow surplus yet forfeit a large option value upon exit, because the irreversible loss includes $\Delta_g(t, \tau, \pi)$—the entire discounted stream of future continuation premia.

Why does this matter? Consider two workers who look identical today—same wage, same current dislike of working. In a static model, they have the same reservation wage and the same welfare cost if a tax hike pushes them out. But in our dynamic model, one might be unclaimed with high survival expectations, giving her a large option value: she stands to gain a lot from staying employed because she accumulates benefits over many future years. The other might be claimed with low survival expectations, so her option value is small. A tax increase that pushes both out destroys far more welfare for the first worker, even though they appeared identical on the surface. The survival-weighted annuity factor $\mu_t(\pi)$ drives this wedge: it enters the option value but plays no role in the static reservation wage.

Irreversibility adds a further asymmetry. When a tax increase pushes workers into retirement, they are gone for good—their option values are permanently destroyed. But when a tax cut retains workers, the benefit is temporary: those workers eventually exit anyway as they age. A tax cut and a tax increase of equal size therefore need not have symmetric welfare effects. The model predicts, in principle, that the welfare cost of a tax restoration can exceed the welfare gain of the preceding cut, even when the magnitudes are identical—a qualitative asymmetry that has no analog in a static model where workers can freely move in and out of employment (Proposition 4).[2]

**Connection to the option value of delay.** The continuation premium $\Delta_g(t, \pi)$ in (22) has a direct analog in the retirement timing literature (cf. Stock and Wise, 1990). In that framework, the option value of delaying retirement depends on the probability that future states of the world make continued work more attractive. Here, higher survival $\sigma_t(\pi)$ raises the expected number of future periods over which favorable disutility draws could arrive, increasing $\Delta_g$. The benefit accrual channel $\lambda_0 \mu_t(\pi)$ provides an additional, unclaimed-specific reason for the option value to be high: each period of continued work raises the effective net wage, and the value of that accrual depends on survival through $\mu_t(\pi)$. The option value framework therefore nests both the standard "waiting for a good draw" motive and the institutional "accrual motive" specific to the Social Security benefit structure.

[2] We do not pursue a quantitative welfare analysis of the 2013 restoration, but this prediction highlights how irreversibility shapes the welfare calculus of temporary tax policies aimed at older workers.

**Exogenous versus endogenous claiming.** The benchmark theory in this section takes claiming status as predetermined. That matches the main empirical design, which compares claimed and unclaimed workers conditional on their observed status. For that purpose, the exogenous-claiming benchmark is the right object: it isolates how the payroll tax holiday changes the effective wedge on work through $\kappa_u$ versus $\kappa_c$, holding the claiming state fixed.

The quantitative implementation is richer. In the counterfactual analysis, initially unclaimed workers may adjust when they claim, so the reported claiming responses should be read as outputs of that richer quantitative block rather than of the benchmark theory alone. A fully endogenous-claiming model would require an expanded state space with benefit entitlement $b_t$, claim status as a transition variable, and different retirement values across claiming states. Without that extra structure, adding a claim-now option directly to the benchmark Bellman problem would risk double counting the value of benefit accrual.

We do not use that richer model as the benchmark for three reasons. First, the policy variation in the payroll tax holiday changes the effective tax rate, not the claiming rules. Second, the SIPP does not observe the full earnings histories and benefit entitlements needed to estimate a joint exit-claiming problem cleanly. Third, the benchmark exogenous-claiming formulation provides the mapping between the theoretical mechanism and the reduced-form evidence.

# 6 Parameterization

This section specifies the functional forms used to take the dynamic retirement model for seniors in Section 5 to data and maps the model's abstract objects to U.S. Social Security institutions. Section 6.1 parameterizes the disutility distribution; Section 6.2 specifies the retirement value; Section 6.3 introduces ex-ante asset heterogeneity; and Sections 6.4.1–6.4.4 map the tax–benefit linkage to institutional rules.

Our guiding principle is simple: each parameter should correspond to one transparent economic force. Some parameters govern how hard work feels as people age or as health worsens. Some govern how attractive retirement is. Others map the Social Security formula into the effective reward from one more year of work. This follows the broad approach of the structural retirement literature, which models retirement as the outcome of age-varying tastes, health, assets, and Social Security incentives (Rust and Phelan, 1997; French, 2005; French and Jones, 2011; Coile and Gruber, 2007; Blundell et al., 2016). The point of our parameterization is therefore not to maximize flexibility. It is to keep the model simple enough

that each estimated object has a clear economic interpretation.

### 6.1 Disutility Distribution

We parameterize the log-disutility mean as a kinked-linear function of age with a health-disutility term:

$$\ln \nu_t \;\sim\; \mathcal{N}\big(m(a,\pi),\ \sigma^2\big), \qquad m(a,\pi) \;=\; m_0 + m_1(a-62) + m_2 \max(0,\ a-65) + \psi\, r_\pi, \quad (41)$$

where $\psi > 0$ is a common health-disutility scale, $r_1 \equiv 1$ normalizes the first survival quartile, and $(r_2, r_3, r_4)$ are health-rank loadings with $1 \geq r_2 \geq r_3 \geq r_4 \geq 0$. Flexible work costs that vary with age and health are standard in structural retirement models (Rust and Phelan, 1997; French, 2005; French and Jones, 2011). We extend by using a kinked-linear age profile after 62 and a scale-rank health decomposition so that the parameters map directly into the density of workers near the retirement margin. The parameter $m_0$ is the baseline level of work disutility at age 62. The parameter $m_1$ is the baseline age slope in work disutility from age 62 onward: if $m_1$ is larger, work becomes harder more quickly as workers age. The parameter $m_2$ is an additional slope that turns on after age 65, so it captures the idea that the disutility of continued work may rise faster once workers are very close to the Full Retirement Age. The health term $\psi r_\pi$ has a similarly simple interpretation. The common scale $\psi$ says how much health matters overall, while the ranks $(r_2, r_3, r_4)$ say which health groups are farther from the healthy benchmark.

The continuation premium $d_{g,t} \equiv V_g^W - V^R$ is denominated in dollars, but the disutility $\nu_t$ is a unitless draw from the log-normal distribution above. To make the two comparable, we divide the dollar premium by a normalization constant $\kappa = 10{,}000$ (Section 7.1.1) before evaluating the continuation probability. The rescaled quantity $d_{g,t}/\kappa$ is now on the same scale as the disutility draws: a worker continues if her realized disutility falls below $d_{g,t}/\kappa$. The value of $\kappa$ acts as an exchange rate between dollars and disutility units—it determines how many dollars of continuation premium are needed to offset one unit of work disutility.

Under this specification, the density condition discussed in the model section is transparent: the log-normal density is unimodal with mode $\bar{\nu}(a,\pi) = \exp(m(a,\pi) - \sigma^2)$, increasing on $(0, \bar{\nu})$ and decreasing on $(\bar{\nu}, \infty)$. When both continuation thresholds lie above the mode, the density is declining, so $\nu_c^* < \nu_u^*$ implies $f(\nu_c^*) \geq f(\nu_u^*)$: claimed workers sit at a denser point.

### 6.2 Retirement Value

The financial part is the Social Security check the worker expects to receive for the rest of her life. This depends on three things: how large her monthly benefit is (her PIA, based on lifetime earnings), whether the benefit is adjusted up or down because she claimed before or after the Full Retirement Age (the multiplier $\phi(a_c)$), and how many years she expects to collect it (the annuity factor $\mu_a^R(\pi)$, which is larger for workers who expect to live longer). The product of these three gives the expected present value of the benefit stream.

The non-financial part captures everything else that makes retirement attractive: more leisure, no commute, time with family. We model this as a constant $\omega_0$ that shifts the value of retirement up or down relative to the financial component. A higher $\omega_0$ means retirement is more appealing even holding benefits fixed. Putting the two parts together, we have

$$V^R(a, a_c, e) \;=\; \underbrace{\text{PIA}(e) \cdot \phi(a_c) \cdot \mu_a^R(\pi)}_{\text{financial: benefit stream}} + \underbrace{\omega_0}_{\text{non-financial: leisure value}}, \tag{42}$$

where $\mu_a^R(\pi) = \sum_{j=1}^{T-a} \beta^j \prod_{k=0}^{j-1} \sigma_{a+k}(\pi)$ is the retirement annuity factor. The non-financial component follows a long tradition in structural retirement models (Stock and Wise, 1990; Gustman and Steinmeier, 1986, 2005; French, 2005).

### 6.3 Ex-Ante Asset Heterogeneity

Assets matter for retirement, but we do not model asset accumulation from youth to old age. Instead, we treat assets as predetermined when the worker enters the model at age 62. This is a simplification. In the broader structural retirement literature, assets are typically a full state variable and retirement choices are made in an incomplete-markets environment (Rust and Phelan, 1997; French, 2005; French and Jones, 2011). Our departure is to compress that dimension into a fixed asset type at age 62. For the short post-62 horizon that we study, the main role of assets is to summarize different starting balance sheets, not to be re-optimized every year.

Formally, each worker enters age 62 with an asset type $x \in \{L, M, H\}$, interpreted as low, middle, and high assets. Asset type is fixed thereafter. This is ex-ante heterogeneity, not a savings choice. Two otherwise similar seniors can face the same tax reform but respond differently because one enters retirement with little liquidity while the other enters with a larger buffer.

The model uses asset type in two ways. First, it shifts the cost of remaining unclaimed:

$$\chi(x) = \chi \cdot s_x, \qquad s_L > s_M > s_H > 0.$$

Low-asset seniors face a larger waiting cost because delaying benefits is harder when liquid resources are scarce. High-asset seniors can afford to wait, so their waiting cost is smaller. This channel governs the strength of claiming pressure.

Second, asset type shifts the distribution of earnings types. Higher-asset seniors are more likely to be in higher lifetime-earnings cells, so asset type changes the probability of drawing a given earnings type $e$. That matters because earnings type determines both the statutory tax base and the marginal benefit accrual rate $\lambda_0(e)$. In the counterfactuals, the asset pattern therefore comes from two forces working together: liquidity differences change how costly it is to delay claiming, and earnings composition changes the size of the effective tax wedge.

## 6.4 Institutional Parameterization

Three institutional components of the U.S. Social Security system determine the effective tax–benefit linkage $\lambda_0(e)$: the PIA formula, the Delayed Retirement Credit, and the Retirement Earnings Test. Mapping these institutional rules into retirement incentives is standard in structural retirement models (Rust and Phelan, 1997; Coile and Gruber, 2007; Blundell et al., 2016). Our contribution is to organize that mapping around the effective wedge, and then use it to connect the reduced-form evidence and the quantitative model. These institutional components matter because they tell us how much of today's payroll tax is really a tax and how much is payment for higher future benefits.

### 6.4.1 Primary Insurance Amount and Earnings Types

The PIA formula converts a worker's Average Indexed Monthly Earnings (AIME) into a monthly benefit amount through a piecewise-linear schedule with three brackets. The bend points are indexed to the year a worker attains age 62. For workers attaining age 62 in 2010, the formula is

$$\text{PIA}(\text{AIME}) = \begin{cases} 0.90 \times \text{AIME} & \text{if AIME} \leq b_1, \\ 0.90 \times b_1 + 0.32 \times (\text{AIME} - b_1) & \text{if } b_1 < \text{AIME} \leq b_2, \\ 0.90 \times b_1 + 0.32 \times (b_2 - b_1) + 0.15 \times (\text{AIME} - b_2) & \text{if AIME} > b_2, \end{cases} \tag{43}$$

where $b_1 = \$761$ and $b_2 = \$4{,}586$ are the 2010 bend points. The PIA formula factors—90%, 32%, and 15%—decline sharply with earnings, so higher-AIME workers receive a smaller fraction of their marginal earnings as benefits.

We discretize the earnings distribution into three types $e \in \{L, M, H\}$ corresponding to the three PIA brackets. The marginal benefit accrual rate for an unclaimed worker of type $e$ is

$$\lambda_0(e) \;=\; r(e) \times \frac{1}{12 \times 35}, \tag{44}$$

where $r(e) \in \{0.90, 0.32, 0.15\}$ is the marginal PIA replacement rate for bracket $e$, and $1/(12 \times 35)$ reflects the conversion from an additional dollar of annual earnings to the change in AIME (monthly average over 35 highest years).[3] Workers above the OASDI taxable maximum ($\$106{,}800$ in 2010–2011) pay no marginal tax and receive no marginal accrual, so $\lambda_0 = 0$ and $\tau^{\text{eff}} = 0$ for earnings above the cap.

The intuition is that the PIA formula determines the marginal reward from one more year of covered earnings. Low earners are on the steep part of the formula, so an extra dollar of earnings translates into a larger future benefit gain. High earners are on the flatter part, so the same extra dollar buys less additional Social Security. This is why the same payroll tax can imply different effective wedges across earnings groups.

#### 6.4.2 Delayed Retirement Credit

Workers who delay claiming benefits past the Full Retirement Age (FRA, age 66 for the 1943–1954 birth cohorts) receive delayed retirement credits equal to $\frac{2}{3}$ of 1 percent per month of delay, or 8 percent per year, up to age 70. For a worker who claims at age $a_{\text{claim}} \geq$ FRA, the monthly benefit is

$$b(a_{\text{claim}}) \;=\; \text{PIA} \times \big[1 + 0.08 \times (a_{\text{claim}} - \text{FRA})\big], \tag{45}$$

where age is measured continuously in years. Under SSA rules the adjustment is applied monthly. Social Security rewards patience. A worker who waits until age 68 to claim—two years past her FRA of 66—receives a monthly check that is 16 percent larger than if she had claimed at 66, and that higher check continues for the rest of her life.

For workers who claim before FRA, benefits are reduced by $\frac{5}{9}$ of 1 percent per month for

[3] This expression assumes that the worker has at least 35 years of positive covered earnings and that current-year earnings are high enough to enter the top-35 set. If either condition fails—e.g., a worker with fewer than 35 positive years displaces a zero, or current earnings fall below all previous top-35 years—the marginal effect on AIME differs from $1/(12 \times 35)$. For workers aged 62 and above with long careers, both conditions are typically satisfied.

the first 36 months of early claiming and $\frac{5}{12}$ of 1 percent per month for any additional months. A worker with FRA 66 who claims at 63 therefore loses $6.67\%$ per year for those three years, but claiming at 62 adds a further reduction at the lower $5\%$-per-year rate. The early reduction schedule is not symmetric with the delayed retirement credit.

#### 6.4.3 Retirement Earnings Test

The Retirement Earnings Test (RET) reduces current benefits for workers who have claimed before FRA and have earnings above an annual exempt amount. For beneficiaries who are under FRA for the entire year, SSA withholds $\$1$ of benefits for every $\$2$ of earnings above the exempt amount ($\$14{,}160$ in 2010; $\theta = 0.50$ in reduced-form notation). In the year a worker attains FRA, the higher exempt amount of $\$37{,}680$ applies, benefits are withheld at $\$1$ for every $\$3$ of excess earnings ($\theta = 1/3$), and only earnings before the month of attaining FRA are tested. After the FRA month, the RET no longer applies. Withheld benefits are partly offset by an upward adjustment of monthly benefits after FRA. SSA recomputes benefits to credit months in which benefits were fully withheld, so the RET is not a pure tax in the usual static sense. However, the present-value offset depends on survival, discounting, and the exact withholding pattern, and for liquidity-constrained or present-biased workers the RET operates as an effective tax on current earnings.

This is the key reason the claimed-worker wedge can be so different from the unclaimed-worker wedge before FRA. Once a worker has claimed, continued work generally does not generate delayed retirement credits, but it can still raise future benefits through recomputation if additional covered earnings replace lower years in the top-35 earnings history. If earnings are also high enough to trigger the RET, current work may nevertheless become less attractive because the worker faces both payroll taxes and temporary withholding of current benefits.

#### 6.4.4 Effective Tax Rate

How much does working actually cost a senior, once we account for all the institutional pieces above? Every worker pays the statutory payroll tax rate $\tau = 0.062$ (the employee OASDI rate). But the effective burden differs. An unclaimed worker earns benefit accrual worth $\lambda_g(e)\,\mu_t(\pi)$ for each dollar of covered earnings, which offsets part of the tax—generously for low earners on the steep part of the PIA formula, and barely at all for high earners on the flat part. A claimed worker who is still below FRA and earning above the RET exempt amount faces the

opposite problem: on top of the payroll tax, SSA withholds part of her current benefit check, adding an effective withholding $\theta$.

Putting these together, the effective tax rate is

$$\tau_g^{\text{eff}}(e,\pi,t) \;=\; \underbrace{\tau}_{\text{statutory tax}} \;-\; \underbrace{\lambda_g(e)\,\mu_t(\pi)}_{\text{accrual offset}} \;+\; \underbrace{\theta\cdot\mathbf{1}\{g=c,\ a_t<\text{FRA},\ w(e)>\bar{y}\}}_{\text{RET for claimed workers}}.$$

During the 2011–2012 payroll tax holiday, $\tau$ fell from $0.062$ to $0.042$. In our calibration, all three earnings types ($\$15{,}000$, $\$40{,}000$, $\$70{,}000$) exceed the RET exempt amount $\bar{y} = \$14{,}160$, so every working claimer below FRA faces the RET withholding.

This equation ties the whole paper together. Congress cut the statutory rate by the same 2 percentage points for everyone, but the effective tax change was not the same. A low-earning unclaimed worker already had a low effective tax because of her generous accrual offset, so the 2pp cut is a smaller proportional change in her effective wedge. A high-earning unclaimed worker gets almost no accrual offset, so her effective tax is close to the statutory rate and the same 2pp cut is proportionally larger. A claimed worker below FRA faces the full statutory tax plus the RET withholding with no accrual offset at all—her effective tax is the highest of the three, and the 2pp cut provides the same dollar relief but operates on a very different baseline. These differences in the effective wedge across earnings types and claiming groups generate the heterogeneous treatment effects that the model matches quantitatively.

# 7 Calibration

We calibrate a dynamic retirement model for seniors to 2010 SIPP employment data—the year immediately before the payroll tax holiday—treating it as the steady state of the benchmark economy. The model begins at age 62 and tracks individuals through retirement. This is not a full lifecycle model. We do not ask how households built their wealth over several decades. Instead, we take the key late-life state variables as given at age 62 and study how they shape work and retirement decisions after that point.

One of those predetermined state variables is assets. We model asset heterogeneity as ex-ante heterogeneity rather than as a post-62 savings problem. This is a reasonable simplification for our setting because most wealth accumulation has already occurred by the early 60s, while our empirical question is about a short-run tax change hitting workers who are already close to retirement.

We estimate the parameters needed to fit retirement behavior. Policy parameters are fixed

at their 2010 statutory values. The remaining non-estimated parameters are either normalization or reduced-form supports introduced because the model does not contain a savings decision or earnings process. Social Security rules therefore follow 2010 law: claiming is permitted from age 62 to 70, the Full Retirement Age is 66, delayed retirement credits accrue after FRA, and the Retirement Earnings Test applies before FRA.

The benchmark economy features a 6.2 percent employee payroll tax. The counterfactual lowers this rate to 4.2 percent, holding everything else fixed. For workers who have not yet claimed Social Security, an additional year of covered earnings generates a future benefit credit. We build this accrual into the dynamic programming problem, which reduces the labor supply incentives relative to workers who have already claimed.

### 7.1 Targeted Moments and Estimation

The model has 11 parameters that we need to estimate: how much workers dislike working at different ages, how health affects the cost of continued employment, and how persistent productivity shocks are. We choose these parameters so that the model's predicted employment patterns match what we observe in the data before the payroll tax holiday.

The data targets are 13 pre-reform employment moments. Nine of them are age-specific employment rates from age 62 to 70, which discipline how quickly workers exit as they age. The remaining four are employment rates for each of the four self-reported health groups (pooled across ages), which discipline how much health matters for the retirement decision. Policy parameters (tax rates, benefit rules, the RET) and the heterogeneity grids (earnings types, asset types) are not estimated. We fix these directly from institutional rules or discretization choices described in Section 6, keeping the estimation focused on the parameters that the data can identify.

Formally, let $m^{\text{data}}$ be the vector of data moments and $m(\theta)$ the model-simulated counterpart. We estimate

$$\hat{\theta} = \arg\min_{\theta} \left[m(\theta) - m^{\text{data}}\right]' W \left[m(\theta) - m^{\text{data}}\right],$$

using a diagonal weight matrix $W$. The age block receives twice the weight of the health block. Within the age block, we place more weight on ages 62–65, where the bulk of retirement transitions occur, and less on ages 66–70, where employment rates are lower and measured less precisely.

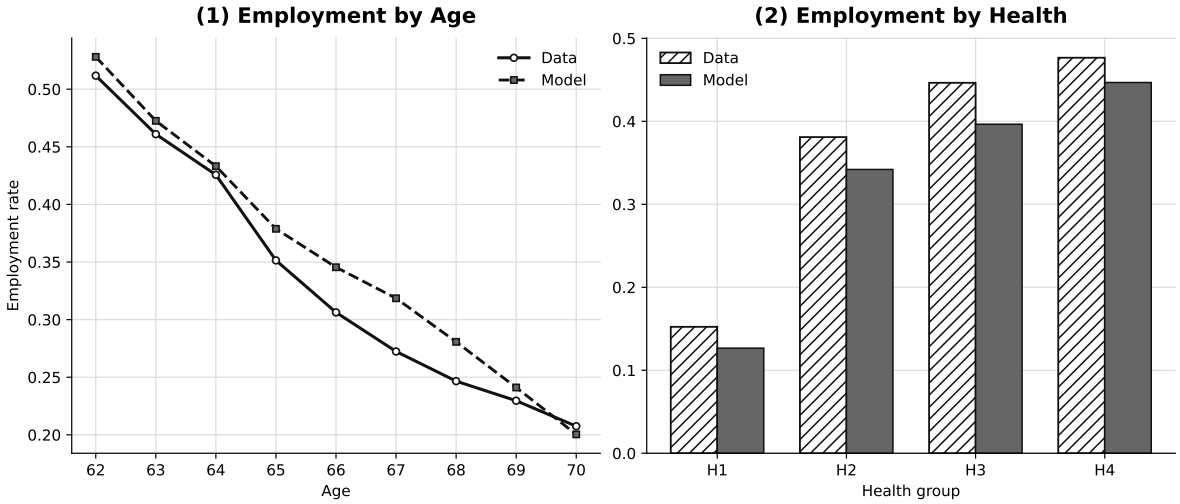



Figure 5: Benchmark fit to targeted moments. Left panel: employment by age, 62–70. Right panel: employment by self-reported health group.

### 7.1.1 Parameters

Table 11 reports the fixed parameters. We group them into three classes: parameters fixed by convention or normalization, parameters fixed by 2010 Social Security law, and parameters that we discipline directly from the data before the SMM. Table 12 reports the 11 behavioral parameters estimated by SMM.

### 7.1.2 Model Fit

Table 13 reports fit to the 13 targeted moments and Figure 5 shows the model fit. The model reproduces the steep drop in employment from age 62 through 64 and tracks the data closely at age 70. It overpredicts employment modestly between ages 65 and 68—the model retires workers somewhat too slowly around the Full Retirement Age. Across health groups, the model understates employment for all four groups, with the largest miss in the second-worst health category $H3$.

Overall, the calibrated model captures the main features of pre-reform senior employment: the rapid retirement between 62 and 65, the modest additional decline after the Full Retirement Age, and the large gap between healthy and unhealthy workers.

### 7.2 Distribution of Effective Wedges

Before turning to employment responses, it is useful to see how much the effective tax varies across workers. We do this with a simple exercise: hold the statutory payroll tax fixed at 6.2 percent for everyone, simulate the model forward, and for each worker still employed at a given age, compute her effective tax rate using (16). The result is a distribution of effective wedges across the workforce at each age. Because we hold the statutory rate fixed, all variation in this distribution comes from Social Security's institutional rules—claiming status, earnings type, survival expectations—rather than from the tax holiday itself.

Figure 6 shows two facts. First, the distribution is wide. The overall mean effective wedge is $-4.4$ percent, with the 10th percentile at $-11.6$ percent, the median at $-9.0$ percent, and the 90th percentile at $6.2$ percent. About 76 percent of active seniors lie below the statutory rate, 17 percent sit exactly at the statutory rate, and 7 percent lie above it. Second, the dispersion is tightly tied to institutional status. Unclaimed workers ages 62–65 face an average wedge of $-8.7$ percent because benefit accrual offsets about 14.9 percentage points of the statutory tax. Claimed workers of the same ages face the highest wedge, 24.2 percent on average, because the RET adds about 18.0 percentage points to the current earnings penalty. After FRA, claimed workers move back to roughly the statutory wedge, while unclaimed workers continue to face negative effective wedges because one more year of work still buys higher future benefits.

The point of the figure is not that workers face different statutory tax rates. They do not. The point is that the same statutory rate maps into very different effective prices of work once Social Security's rules are taken seriously. That distribution of effective wedges is the object that the reduced-form evidence and the counterfactual exercises are both trying to discipline.

### 7.3 Results

Who responds to the tax cut, and who does not? Table 14 breaks down the employment response across age, health, assets, and earnings. The answer reveals a clear pattern: the tax cut moves workers who are close to the retirement margin but have not yet crossed it.

*Age.* Employment rises by 1.1 percentage points at ages 62–65 and by 1.3 points at ages 66–70. If anything, the effect is slightly larger at older ages, where more workers are on the margin of exit.

*Health.* Here the gradient is striking. Workers in the best health group ($H4$) show the largest gain (1.9 pp), while those in the worst group ($H1$) barely respond at all (0.0 pp). The reason is selection: the sickest workers have mostly already retired, so there are few of them

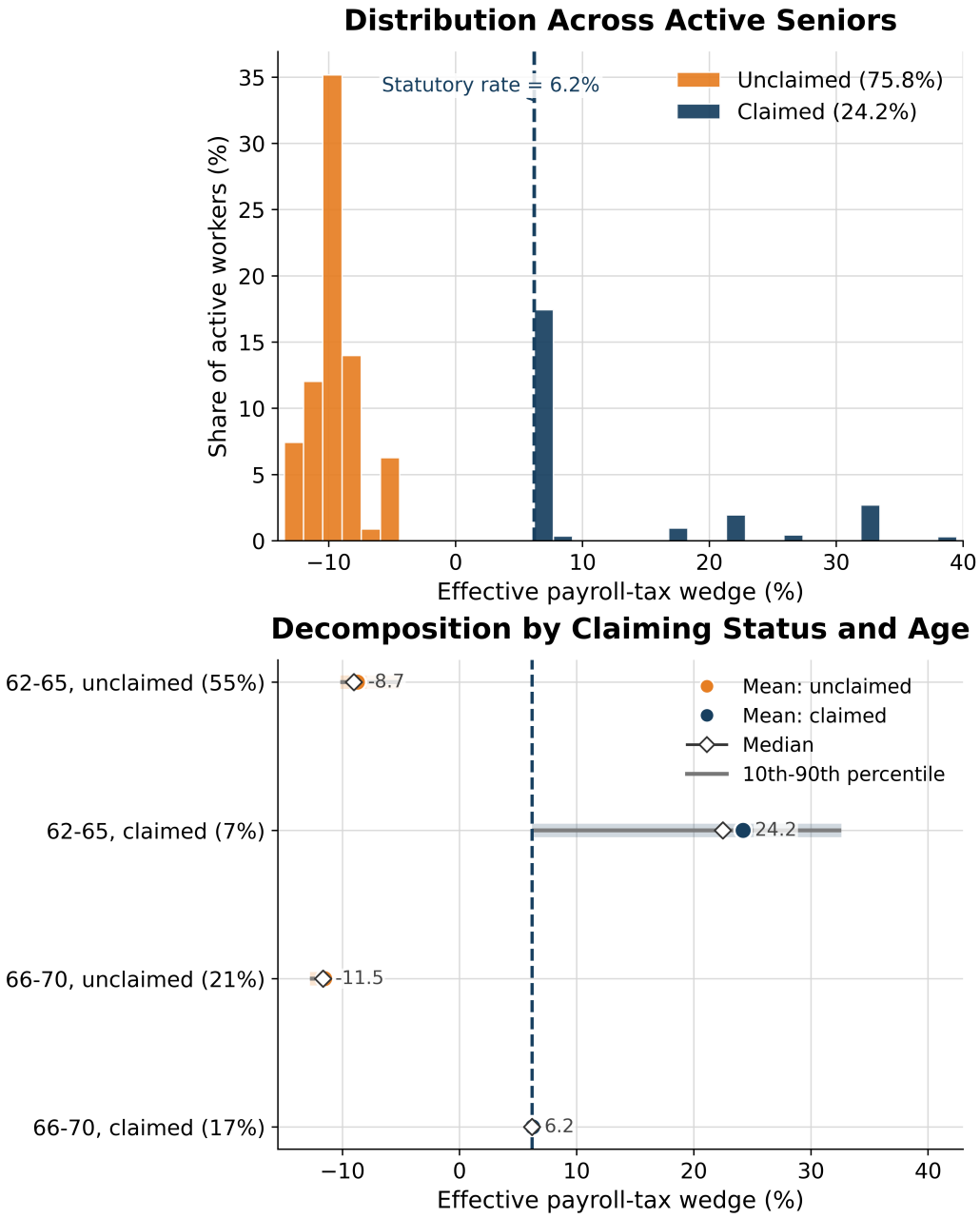



Figure 6: Model-implied distribution of effective payroll-tax wedges. Panel A plots the weighted histogram of effective wedges among active workers ages 62–70 under a constant 6.2 percent employee payroll tax. Panel B decomposes the distribution by claiming status and age block; dots are means, hollow markers are medians, and horizontal bars show the 10th to 90th percentile range. The statutory payroll tax rate is shown by the dashed vertical line.

left near the margin for the tax cut to reach. The healthiest workers are the ones still working and still deciding whether to stay—exactly the population a tax cut can affect.

*Assets and earnings.* Among seniors who are still working when the tax holiday arrives, low-asset workers respond the most (1.5 pp), compared with 1.1 pp for middle- and high-asset groups. This does not contradict the health gradient above. Poor health mostly works through selection out of employment before the reform, while low assets matter within the remaining pool of workers by making current earnings more valuable and delaying retirement harder to self-finance. Along the earnings dimension, the response peaks at low earners ($30k, 1.2 pp) and flattens at higher levels. High earners are closer to the taxable maximum, which compresses the effective tax change they experience.

What drives these patterns? To find out, Table 15 decomposes the response by shutting off heterogeneity channels. In each panel, we start from a stripped-down model in which every group has a common baseline response, and then add channels back to see which ones pull the groups apart.

**Assets (Panel A).** Without asset-dependent channels, every group would respond by 2.1 percentage points. Two channels pull groups apart. The waiting-cost channel captures how assets change the cost of delaying claiming: low-asset households face strong claiming pressure, leaving less room for the tax cut to raise employment; middle- and high-asset households face weaker claiming pressure, which redirects part of the reform to the claiming margin. The earnings composition channel captures the correlation between assets and earnings types: high-asset households are likely to be higher wage types, which raises their response, but this force is largely offset by the interaction with the claiming channel. On net, asset heterogeneity compresses the average response from the 2.1-point baseline to about 1.1–1.5 points.

**Health (Panel B).** If we shut off all health-specific channels, every health group would respond by the same 0.61 percentage points. What creates the large spread from $H1$ to $H4$? Not longevity differences alone—those turn out to be too small to matter much. The two channels that do the heavy lifting are work capacity and earnings composition. Unhealthy workers find it physically harder to keep working and face more frequent negative shocks, which pushes them toward retirement regardless of the tax rate. At the same time, healthier workers tend to hold higher-paying jobs, which raises their effective tax change. Together, these two forces account for nearly all of the health gradient. The sickest group barely responds because the

costs of continued work are so high that a 2pp tax cut is not enough to keep them employed.

**Interpretation.** The two decompositions tell a simple story. Assets and health play different roles: assets determine how much a worker needs the extra income from continued work, while health determines whether she is physically able to supply it. A low-asset worker in good health is the most responsive type—she is near the retirement margin, able to keep working, and sensitive to the effective price of labor. A high-asset worker in poor health is the least responsive—she has little financial need to work and high costs of doing so.

The broader lesson is that the aggregate employment effect of a tax reform depends not on the average incentive it creates, but on where in the joint distribution of wealth, earnings, and health the marginal workers sit. Two policies with identical fiscal cost can produce very different responses if they reach different parts of this distribution.

# 8 Counterfactual Exercises

We use the calibrated model to study four policy experiments. The first is a permanent cut in the payroll tax rate from 6.2 to 4.2 percent. This is deliberately stronger than the historical two-year holiday; interpret it as a comparative-static experiment that isolates the incentive channel. The second removes the Retirement Earnings Test (RET) while holding all other policy rules fixed. The third shuts down the marginal link between post-62 earnings and future benefits while holding benefit levels fixed. The fourth aggregates overlapping cohorts and imposes the actual 2011–2012 holiday as an unexpected calendar-year shock. Because assets are predetermined at age 62 in our model, differences across asset groups below should be read as differences across seniors who start retirement with different balance sheets, not as the result of a new savings response after the reform.

## 8.1 Substitution Versus Income Effects

In the model, cutting the payroll tax raises employment for ages 62–70 from 0.358 to 0.370, or by about 1.2 percentage points. The basic question is why this happens. Is it because seniors simply have more money, or because working becomes more rewarding?

To answer that question, we compare the tax cut to two simpler cases. First, we replace the tax cut with a lump-sum transfer that costs the government the same amount. This transfer lowers employment slightly, from 0.358 to 0.357. In other words, giving seniors extra income by itself does not make them work more; if anything, it makes them retire a bit earlier.

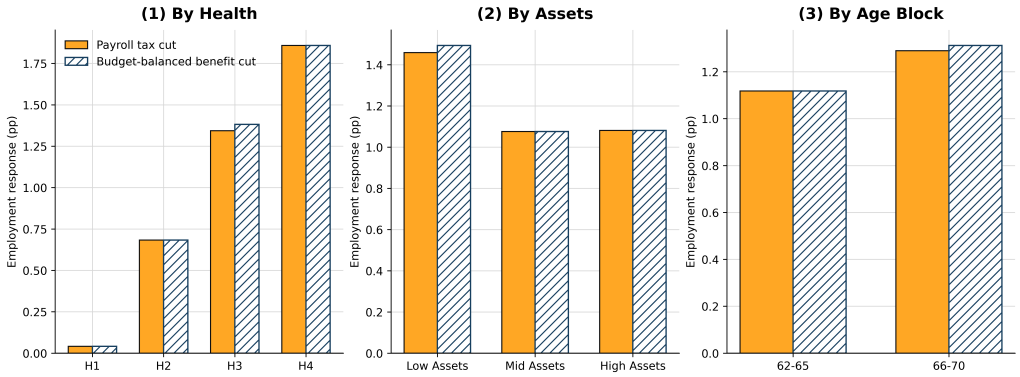



Figure 7: Payroll-tax cut versus budget-balanced reform. Bars report employment responses by health, assets, and age block. The budget-balanced reform finances the tax cut with a proportional benefit reduction chosen to match the benchmark fiscal balance over ages 62–90.

Second, we require the reform to be budget balanced. We let the payroll-tax cut be financed by a small proportional cut in Social Security benefits. The model implies that the required benefit cut is only about 0.86 percent. Even with this financing, employment is still 0.3704, which is almost the same as in the original tax-cut experiment.

The lesson is straightforward. The payroll-tax reform matters mainly because it changes the payoff from working one more year. That is a substitution effect. A pure income transfer does not generate the same response, and adding a small benefit cut barely changes the result. Figure 7 confirms this pattern across health, asset, and age groups: the tax-cut and budget-balanced bars are nearly identical in every subgroup. So the main channel in the model is the incentive to work, not the extra cash itself.

## 8.2 Removing the Retirement Earnings Test

The second and third exercises isolate two distinct channels through which Social Security shapes senior labor supply. We begin with the Retirement Earnings Test. Before the Full Retirement Age, a worker who has already claimed benefits loses $1 of benefits for every $2 earned above the exempt amount. On paper, this is a timing device: withheld benefits are mostly returned later through a higher monthly payment. For seniors with enough savings and a long life expectancy, the repayment is close to actuarially fair. But for early claimers with little savings, a dollar returned years later is worth much less than a dollar lost today. For them, the RET is effectively a tax that penalizes combining work with benefit receipt before FRA.

We run this exercise for two reasons. First, it is directly policy-relevant. Congress eliminated the RET for workers above the FRA in the Senior Citizens' Freedom to Work Act of 2000, but kept it in place for workers between 62 and the FRA—precisely the age range in our model. Whether to extend the repeal to pre-FRA workers has been raised in subsequent legislative proposals. Second, it helps interpret our main mechanism. In the data and in the model, claimed workers respond differently from unclaimed workers. That gap can arise for two reasons: claimed workers no longer earn delayed retirement credits from postponing claiming, and before FRA they may also face the RET. Shutting down the RET isolates the second force.

In the counterfactual, we set the RET withholding rates to zero and leave all other parameters unchanged. It asks what happens if seniors who already claimed can keep their full current benefit while continuing to work before FRA. The result is large. Table 6 reports the main moments. Employment at ages 62–65 rises from 45.4 to 58.3 percent, a 12.9 percentage-point increase. The effect is much smaller after FRA, where the RET no longer applies mechanically: employment at ages 66–70 rises by 4.0 percentage points. Claiming also rises before FRA, from 58.1 to 64.4 percent. The reason is intuitive. Once the RET penalty is removed, some seniors no longer have to choose between claiming for current income and continuing to work.

The sharpest pattern is across assets. Pre-FRA employment rises by 18.1 percentage points for the low-asset group, 14.6 points for the middle-asset group, and only 3.0 points for the high-asset group. Low-asset seniors value current benefit income the most, so the RET is especially distortionary for them. Removing it lets them claim for liquidity reasons without leaving work at the same time.

Two findings stand out. First, the RET is quantitatively large: removing it raises pre-FRA employment by 12.9 percentage points, about ten times the 1.2 percentage-point effect of the permanent payroll-tax cut. Second, the response is concentrated among low-asset seniors, the group for whom current benefit income matters most. This implies that the RET does not simply shift the timing of benefit receipt—it actively discourages work among seniors who need current income the most.

### 8.3 Delinking Benefits from Additional Earnings

The delinking counterfactual works through a different channel. It applies to workers who have not yet claimed. For them, working longer can raise future Social Security benefits, so continued work brings both current wages and higher future benefits. If we remove that link, working longer becomes less attractive, and claiming begins earlier. This is not the same

Table 6: Removing the Retirement Earnings Test: Employment and Claiming Responses

| | Employment rate (%) | | | Claiming rate (%) | | |
|---|---|---|---|---|---|---|
| | Benchmark | No RET | Diff. | Benchmark | No RET | Diff. |
| *Panel A. Overall* | | | | | | |
| Ages 62–65 | 45.4 | 58.3 | 12.9 | 58.1 | 64.4 | 6.3 |
| Ages 66–70 | 27.8 | 31.8 | 4.0 | 80.9 | 81.6 | 0.7 |
| Ages 62–70 | 35.8 | 43.9 | 8.0 | 70.5 | 73.8 | 3.3 |
| *Panel B. Pre-FRA employment by asset group (ages 62–65)* | | | | | | |
| | Employment rate (%) | | | | | |
| | Benchmark | No RET | Diff. | | | |
| Low assets | 37.0 | 55.2 | 18.1 | | | |
| Mid assets | 44.1 | 58.6 | 14.6 | | | |
| High assets | 59.1 | 62.1 | 3.0 | | | |

Notes: Panel A reports benchmark and no-RET employment and claiming rates. Panel B reports benchmark and no-RET employment rates by ex-ante asset group for ages 62–65. The no-RET counterfactual sets the Retirement Earnings Test withholding rates to zero while holding all other calibrated parameters fixed. Differences are in percentage points.

mechanism as the RET. The RET changes the return to working *after* claiming. Benefit accrual changes the return to working *before* claiming.

The RET exercise isolates one part of the claimed-worker wedge. A different question is what happens if the Social Security system stops rewarding extra work after age 62 through higher future benefits. This is the most direct counterfactual for our mechanism because the paper's central object is benefit linkage.

It is important to be precise about what we change. We do *not* give everyone the same Social Security benefit. That would be a much broader redistribution reform. Instead, we hold benefit levels fixed and remove only the extra benefit credit from one more year of work after age 62. In other words, we ask what happens if older workers no longer earn additional future benefits by staying on the job.

The answer is clear. Table 7 shows that delinking lowers employment and raises claiming. Employment at ages 62–65 falls from 45.4 to 41.4 percent, a 4.0 percentage-point decline. Over ages 62–70, employment falls by 2.1 percentage points, while claiming rises by 4.3 percentage points. The largest employment decline occurs right at age 62, where employment falls by about 5.0 percentage points. The basic intuition is simple: when extra work no longer raises future benefits, the payoff to continued work falls, so some seniors claim earlier and leave work sooner.

The heterogeneity is also informative. The pre-FRA employment decline is essentially

zero for the lowest-asset group, 2.2 percentage points for the middle-asset group, and 12.5 percentage points for the highest-asset group. This does not mean high-asset seniors are worse off in a welfare sense. It means they are the group with the most room to adjust. In the benchmark, low-asset seniors already tend to claim early, so there is little additional margin to lose. By contrast, higher-asset seniors are the ones who use the option to delay claiming and keep working. Once that extra reward from work is removed, they move noticeably toward earlier claiming and earlier exit. The same logic explains why the effect is strongest for healthy seniors, who have the most scope to delay retirement in the benchmark.

This is the key takeaway. In the model, benefit linkage sustains late-life work by rewarding continued earnings among seniors who still have room to delay claiming. Removing that linkage pushes claiming earlier and lowers employment.

Table 7: Delinking Post-62 Benefits from Additional Earnings

| | Employment rate (%) | | | Claiming rate (%) | | |
|---|---|---|---|---|---|---|
| | Benchmark | Delinked | Diff. | Benchmark | Delinked | Diff. |
| *Panel A. Overall* | | | | | | |
| Ages 62–65 | 45.4 | 41.4 | -4.0 | 58.1 | 61.3 | 3.2 |
| Ages 66–70 | 27.8 | 27.3 | -0.5 | 80.9 | 86.2 | 5.3 |
| Ages 62–70 | 35.8 | 33.7 | -2.1 | 70.5 | 74.8 | 4.3 |
| *Panel B. Pre-FRA employment by asset group (ages 62–65)* | | | | | | |
| | Employment rate (%) | | | | | |
| | Benchmark | Delinked | Diff. | | | |
| low_assets | 37.0 | 37.0 | 0.0 | | | |
| mid_assets | 44.1 | 41.9 | -2.2 | | | |
| high_assets | 59.1 | 46.7 | -12.5 | | | |

Notes: The counterfactual sets post-62 benefit accrual from additional earnings to zero while holding the level of Social Security benefits and all other calibrated parameters fixed. Differences are in percentage points.

### 8.4 Unexpected Holiday Shock as MIT Shock

The natural counterpart to the policy episode is an unexpected payroll-tax shock. We therefore build a calendar-year sequence with overlapping cohorts. Each year, a new equal-mass cohort enters the model at age 62. Cohorts already in the model follow the no-holiday benchmark through 2010 and are hit by the holiday unexpectedly at the start of 2011. Cohorts entering in 2011 and 2012 face the lower tax at entry. We then aggregate outcomes over ages 62–70 within each calendar year, matching the reduced-form sample more closely than the

single-cohort exercise.

We model the 2011 payroll-tax holiday as an MIT shock. Before 2011, agents do not anticipate the reform and all cohorts evolve under the no-holiday benchmark with a constant 6.2 percent employee payroll tax. At the start of 2011, the tax path switches unexpectedly to 4.2 percent in 2011 and 2012 and then returns to 6.2 percent from 2013 onward; once the shock arrives, agents correctly foresee this temporary path. For cohorts already in the model in 2011, we take the full benchmark state distribution at the end of 2010, including employment status, claiming status, health, assets, preferences, and idiosyncratic shocks, and use it as the initial condition for the post-shock continuation problem. Cohorts that enter at age 62 in 2011 or 2012 face the lower tax upon entry. We then aggregate outcomes across overlapping cohorts ages 62 to 70 within each calendar year. This construction removes anticipatory responses before 2011 and makes the model counterpart line up with the calendar-year policy timing in the data.

**Computational algorithm.** The exercise is a direct nonlinear transition calculation. For each cohort-entry year, we proceed in five steps. First, we solve the benchmark dynamic program by backward induction under the no-holiday path, with a constant 6.2 percent employee payroll tax. Second, we solve the shocked dynamic program under the temporary holiday path for that cohort's calendar years: 4.2 percent in 2011 and 2012 and 6.2 percent otherwise. Third, we simulate the benchmark economy forward. If the cohort is already alive in 2011, we stop the benchmark simulation at the end of 2010 and record the full cross-sectional state, including active unclaimed, active claimed, retired unclaimed, and retired claimed masses by health, asset, preference, and wage-shock states. Fourth, we use that benchmark state as the initial condition for the shocked continuation problem starting in 2011. Cohorts that enter in 2011 or later simply enter directly under the shocked policy path. Fifth, we aggregate benchmark and shocked outcomes across equal-mass overlapping cohorts and compute calendar-year employment for ages 62–70. This algorithm makes the MIT timing explicit: the reform is unanticipated before 2011, but once it arrives agents understand the full temporary path.

**Results.** Figure 8 displays the result. Panel (1) plots the employment stock under the no-holiday benchmark (flat at about 35.8 percent) against the holiday path. During the holiday years, employment rises sharply—from 35.8 to 36.2 percent in 2011 and 36.3 percent in 2012—as new age-62 entrants face a lower tax at the main retirement margin. Panel (2) shows the

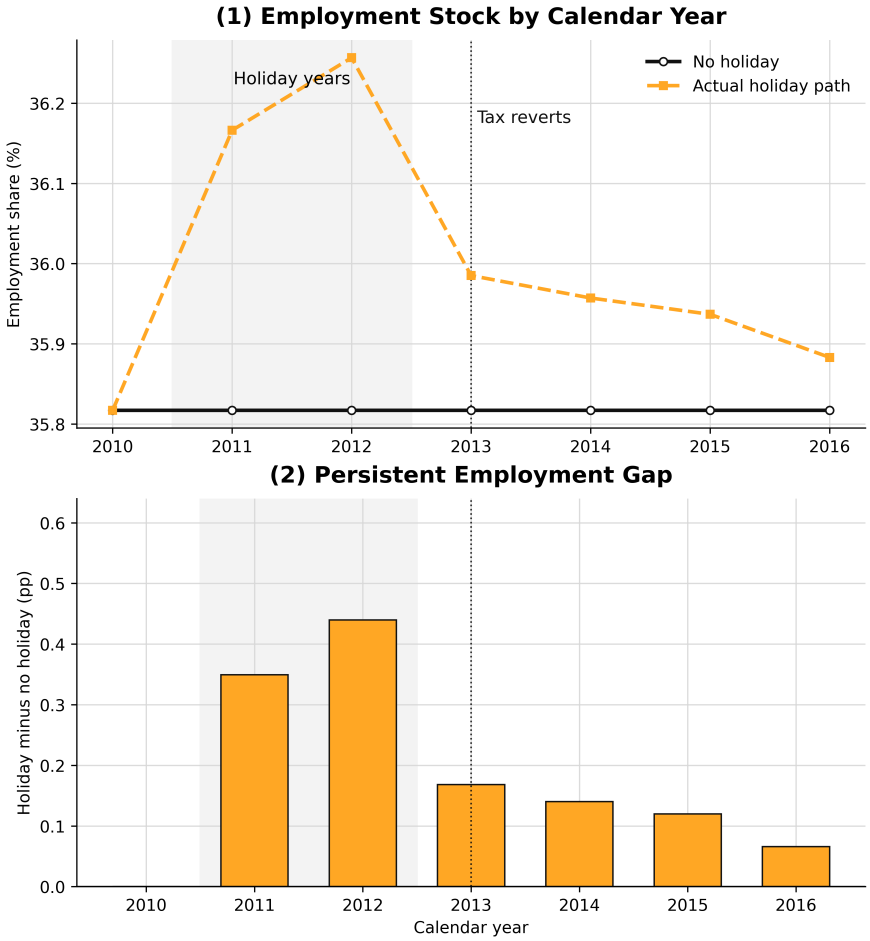



Figure 8: Unexpected 2011–2012 holiday with overlapping cohorts. Each calendar year aggregates equal-mass cohorts ages 62–70. Panel (1) compares the employment path under no holiday and under the actual holiday with reversion in 2013. Panel (2) plots the resulting employment gap. Because retirement is absorbing, the gap remains positive after the tax reverts.

corresponding gap: 0.35 percentage points in 2011, rising to 0.44 points in 2012.

The key feature is what happens after the tax reverts in 2013. The gap drops immediately—from 0.44 to 0.17 percentage points—but it does not vanish. Employment remains above the no-holiday path through 2016, declining gradually from 0.17 to 0.07 percentage points. This is exactly the stock-gap persistence from Section 5. Retirement is irreversible: once a senior exits, she does not return. The holiday prevents some exits in 2011–2012, building a larger employment stock. When the tax reverts in 2013, no new retention occurs, but the seniors who were kept in work during the holiday remain employed.

# 9 Conclusion

This paper asks how much Social Security transforms a common payroll-tax reform into different effective work incentives for older workers, and whether those differences matter for retirement behavior. For seniors, the statutory payroll-tax rate is not the right policy object. What matters is the effective wedge on work after benefit accrual, claiming status, and the Retirement Earnings Test are taken into account. Once these institutional features are incorporated, a common payroll-tax reform no longer implies a common incentive change.

Using the 2011–2012 payroll-tax holiday and monthly SIPP data on workers ages 62–70, we show that retirement responses vary systematically with baseline tax exposure, claiming status, and the strength of marginal benefit accrual—exactly the margins along which Social Security creates dispersion in effective wedges. A dynamic retirement model organized around the same effective-wedge object quantifies the policy margins behind this heterogeneity: a permanent payroll-tax cut works mainly through substitution, removing the RET raises employment sharply among low-asset early claimers, shutting down benefit-accrual lowers employment among seniors with room to delay, and because retirement is absorbing, employment gains from the temporary holiday persist after the tax reverts.

This paper's contribution is therefore not to show that Social Security matters for retirement in general which is already well known. The contribution is to show that a common payroll-tax reform has heterogeneous incidence across seniors because Social Security itself converts the same statutory change into different effective wedges. That shift in perspective matters for both empirical work and policy analysis. A single senior labor-supply elasticity is not enough when the relevant object is a distribution of wedges. For older workers, payroll-tax policy cannot be evaluated from the statutory rate alone. It must be evaluated through the institutional mapping from the statutory rate to the effective return to work.

# A Omitted Proofs

**Assumption 2** (Regularity for monotone comparative statics)**.** *The conditional law of $\nu_{t+1}$ under $\mathbb{E}_t[\cdot]$ is independent of $(g, \pi, \tau)$. In addition, $V^R(t) \geq 0$ for every $t$.*

*Proof.* Throughout the proof, write

$$x_+ \equiv \max\{x, 0\}.$$

By (21), (23), and (22), the continuation branch in the Bellman equation can be written as

$$w(1-\tau) + A_g(t,\pi) - \nu_t + \beta\sigma_t(\pi)\mathbb{E}_t\big[V_g^W(t+1, \nu_{t+1}, \pi; \tau)\big] = V^R(t) + \nu_g^*(t,\tau,\pi) - \nu_t.$$

Hence, for every $t \in \{0, \ldots, N-1\}$,

$$V_g^W(t, \nu_t, \pi; \tau) = V^R(t) + \big(\nu_g^*(t,\tau,\pi) - \nu_t\big)_+. \tag{46}$$

We use (46) repeatedly.

*Part (i).* Define the threshold gap

$$G_t(\pi;\tau) \equiv \nu_u^*(t,\tau,\pi) - \nu_c^*(t,\tau,\pi).$$

At $t = N-1$, the terminal condition $V_u^W(N, \nu_N, \pi; \tau) = V_c^W(N, \nu_N, \pi; \tau) = V^R(N)$ implies

$$G_{N-1}(\pi;\tau) = A_u(N-1,\pi) - A_c(N-1,\pi).$$

Using (17),

$$A_u(t,\pi) - A_c(t,\pi) = w\big[\kappa_u(t,\pi) - \kappa_c(t,\pi)\big] = w\Big[\lambda_0(e)\mu_t(\pi) + \theta\,\mathbf{1}\{a_t < \text{FRA},\ w(e) > \bar{y}\}\Big] > 0,$$

so $G_{N-1}(\pi;\tau) > 0$.

Now suppose $G_{t+1}(\pi;\tau) > 0$ for some $t \leq N-2$. By (46), for every realization of $\nu_{t+1}$,

$$V_u^W(t+1, \nu_{t+1}, \pi; \tau) - V_c^W(t+1, \nu_{t+1}, \pi; \tau) = \big(\nu_u^*(t+1,\tau,\pi) - \nu_{t+1}\big)_+ - \big(\nu_c^*(t+1,\tau,\pi) - \nu_{t+1}\big)_+ \geq 0,$$

because $\nu_u^*(t+1,\tau,\pi) > \nu_c^*(t+1,\tau,\pi)$. Therefore,

$$\mathbb{E}_t\big[V_u^W(t+1, \nu_{t+1}, \pi; \tau) - V_c^W(t+1, \nu_{t+1}, \pi; \tau)\big] \geq 0.$$

Using (22),

$$G_t(\pi;\tau) = \big[A_u(t,\pi) - A_c(t,\pi)\big] + \beta\sigma_t(\pi)\mathbb{E}_t\big[V_u^W(t+1,\nu_{t+1},\pi;\tau) - V_c^W(t+1,\nu_{t+1},\pi;\tau)\big] > 0.$$

Backward induction yields $G_t(\pi;\tau) > 0$ for every $t \in \{0,\dots,N-1\}$, which is (24). Finally, (46) implies the pointwise weak value ordering

$$V_u^W(t,\nu_t,\pi;\tau) \geq V_c^W(t,\nu_t,\pi;\tau) \qquad \text{for all } (t,\nu_t,\pi,\tau).$$

Thus unclaimed workers tolerate strictly higher work disutility before exiting.

*Part (ii): monotonicity of each threshold in survival type.* Fix $g \in \{c,u\}$ and $\pi' \geq \pi$. We prove by backward induction that

$$\nu_g^*(t,\tau,\pi') \geq \nu_g^*(t,\tau,\pi) \qquad \text{for every } t.$$

Because each survival probability $\sigma_{t+k}(\pi)$ is weakly increasing in $\pi$, every summand in (13) is weakly increasing in $\pi$, so $\mu_t(\pi)$ is weakly increasing in $\pi$. Hence $A_g(t,\pi)$ is weakly increasing in $\pi$ for each $g$.

At $t = N-1$,

$$\nu_g^*(N-1,\tau,\pi') - \nu_g^*(N-1,\tau,\pi) = \big[A_g(N-1,\pi') - A_g(N-1,\pi)\big] + \beta\big[\sigma_{N-1}(\pi') - \sigma_{N-1}(\pi)\big]V^R(N) \geq 0,$$

where the inequality uses $A_g(N-1,\pi') \geq A_g(N-1,\pi)$ and Assumption 2.

Now suppose

$$\nu_g^*(t+1,\tau,\pi') \geq \nu_g^*(t+1,\tau,\pi)$$

for some $t \leq N-2$. By (46), for every realization of $\nu_{t+1}$,

$$V_g^W(t+1,\nu_{t+1},\pi';\tau) - V_g^W(t+1,\nu_{t+1},\pi;\tau) = \big(\nu_g^*(t+1,\tau,\pi') - \nu_{t+1}\big)_+ - \big(\nu_g^*(t+1,\tau,\pi) - \nu_{t+1}\big)_+ \geq 0.$$

Since the law of $\nu_{t+1}$ is independent of $\pi$ by Assumption 2,

$$\Psi_g(t,\pi';\tau) \equiv \mathbb{E}_t\big[V_g^W(t+1,\nu_{t+1},\pi';\tau)\big] \geq \mathbb{E}_t\big[V_g^W(t+1,\nu_{t+1},\pi;\tau)\big] \equiv \Psi_g(t,\pi;\tau).$$

Moreover, (46) and Assumption 2 imply $\Psi_g(t,\pi';\tau) \geq V^R(t+1) \geq 0$. Therefore,

$$\sigma_t(\pi')\Psi_g(t,\pi';\tau) - \sigma_t(\pi)\Psi_g(t,\pi;\tau) = \big[\sigma_t(\pi') - \sigma_t(\pi)\big]\Psi_g(t,\pi';\tau) + \sigma_t(\pi)\big[\Psi_g(t,\pi';\tau) - \Psi_g(t,\pi;\tau)\big] \geq 0.$$

Combining this with $A_g(t,\pi') \geq A_g(t,\pi)$ in (22) gives

$$\nu_g^*(t,\tau,\pi') \geq \nu_g^*(t,\tau,\pi).$$

This proves (25).

*Part (ii): monotonicity of the threshold gap in survival type.* We next show that $G_t(\pi;\tau)$ is weakly increasing in $\pi$. At $t = N-1$,

$$G_{N-1}(\pi;\tau) = A_u(N-1,\pi) - A_c(N-1,\pi) = w\Big[\lambda_0(e)\mu_{N-1}(\pi) + \theta\,\mathbf{1}\{a_{N-1} < \text{FRA},\ w(e) > \bar{y}\}\Big],$$

which is weakly increasing in $\pi$ because $\mu_{N-1}(\pi)$ is.

Now fix $t \leq N-2$ and assume that at date $t+1$ both thresholds are weakly increasing in $\pi$ and the gap $G_{t+1}(\pi;\tau)$ is weakly increasing in $\pi$. For a fixed realization of $\nu_{t+1}$, define

$$D_{t+1}(\nu_{t+1},\pi;\tau) \equiv V_u^W(t+1,\nu_{t+1},\pi;\tau) - V_c^W(t+1,\nu_{t+1},\pi;\tau).$$

By (46),

$$D_{t+1}(\nu_{t+1},\pi;\tau) = \big(\nu_u^*(t+1,\tau,\pi) - \nu_{t+1}\big)_+ - \big(\nu_c^*(t+1,\tau,\pi) - \nu_{t+1}\big)_+.$$

Using part (i), this can be rewritten as

$$D_{t+1}(\nu_{t+1},\pi;\tau) = \begin{cases} G_{t+1}(\pi;\tau), & \nu_{t+1} < \nu_c^*(t+1,\tau,\pi), \\ \nu_u^*(t+1,\tau,\pi) - \nu_{t+1}, & \nu_c^*(t+1,\tau,\pi) \leq \nu_{t+1} < \nu_u^*(t+1,\tau,\pi), \\ 0, & \nu_{t+1} \geq \nu_u^*(t+1,\tau,\pi). \end{cases}$$

Because $\nu_u^*(t+1,\tau,\pi)$ and $G_{t+1}(\pi;\tau)$ are weakly increasing in $\pi$, a direct check of the three regions shows that $D_{t+1}(\nu_{t+1},\pi;\tau)$ is weakly increasing in $\pi$ for every fixed $\nu_{t+1}$. Since $D_{t+1}(\nu_{t+1},\pi;\tau) \geq 0$ by part (i), Assumption 2 gives

$$\mathbb{E}_t\big[D_{t+1}(\nu_{t+1},\pi';\tau)\big] \geq \mathbb{E}_t\big[D_{t+1}(\nu_{t+1},\pi;\tau)\big] \geq 0.$$

Therefore,

$$\begin{aligned} G_t(\pi';\tau) - G_t(\pi;\tau) &= \big[A_u(t,\pi') - A_c(t,\pi')\big] - \big[A_u(t,\pi) - A_c(t,\pi)\big] \\ &\quad + \beta\Big(\sigma_t(\pi')\mathbb{E}_t\big[D_{t+1}(\nu_{t+1},\pi';\tau)\big] - \sigma_t(\pi)\mathbb{E}_t\big[D_{t+1}(\nu_{t+1},\pi;\tau)\big]\Big) \\ &\geq 0, \end{aligned}$$

where the last inequality uses the monotonicity of $\mu_t(\pi)$, the monotonicity of $D_{t+1}$ just established, and $\sigma_t(\pi') \geq \sigma_t(\pi)$. This proves (26).

*Part (iii).* Fix $g \in \{c, u\}$ and $\tau' \geq \tau$. We prove by backward induction that

$$\nu_g^*(t,\tau',\pi) \leq \nu_g^*(t,\tau,\pi) \qquad \text{for every } t.$$

At $t = N-1$,

$$\nu_g^*(N-1,\tau',\pi) - \nu_g^*(N-1,\tau,\pi) = -w(\tau'-\tau) \leq 0.$$

Now suppose $\nu_g^*(t+1,\tau',\pi) \leq \nu_g^*(t+1,\tau,\pi)$ for some $t \leq N-2$. By (46), for every realization of $\nu_{t+1}$,

$$V_g^W(t+1,\nu_{t+1},\pi;\tau') \leq V_g^W(t+1,\nu_{t+1},\pi;\tau).$$

Since the law of $\nu_{t+1}$ is independent of $\tau$ by Assumption 2,

$$\Delta_g(t,\tau',\pi) = \beta\sigma_t(\pi)\mathbb{E}_t\big[V_g^W(t{+}1,\nu_{t+1},\pi;\tau')\big] - V^R(t) \leq \beta\sigma_t(\pi)\mathbb{E}_t\big[V_g^W(t{+}1,\nu_{t+1},\pi;\tau)\big] - V^R(t) = \Delta_g(t,\tau,\pi).$$

Substituting into (22) gives

$$\nu_g^*(t,\tau',\pi) - \nu_g^*(t,\tau,\pi) = -w(\tau'-\tau) + \Delta_g(t,\tau',\pi) - \Delta_g(t,\tau,\pi) \leq 0.$$

Backward induction proves (27). □

# B SIPP Robustness and Supplemental Evidence

This appendix collects all supplementary diagnostics and robustness checks referenced in Section 4.

## B.1 Conditional Parallel-Trends Diagnostics

The main SIPP analysis uses three common-shock exposure designs. In each case, the holiday starts at the same calendar date for everyone, and identification comes from predetermined differences in exposure. Figures 9 to 11 report the corresponding timing diagnostics. For the tax-exposure design, the cleanest diagnostic is the monthly coefficient event study from the regression itself. For the claiming-status and $\lambda_0$ designs, the appendix shows treatment and comparison group paths, with the $\lambda_0$ companion aggregated to quarterly frequency to make the underlying trend easier to see. Because the estimation window begins in January 2010, the pre-period is short, so these figures are diagnostics rather than formal proof. They are meant to show whether the relevant groups were moving similarly before the holiday and how those paths diverged after the reform began in January 2011.

The first and third designs show broadly similar pre-period paths. The claiming-status comparison is less flat in 2010, so that design should be read together with the rest of the evidence rather than in isolation. That is one reason the paper emphasizes the full pattern across the three designs, especially the continuous-$\lambda_0$ result, instead of leaning on any single coefficient.

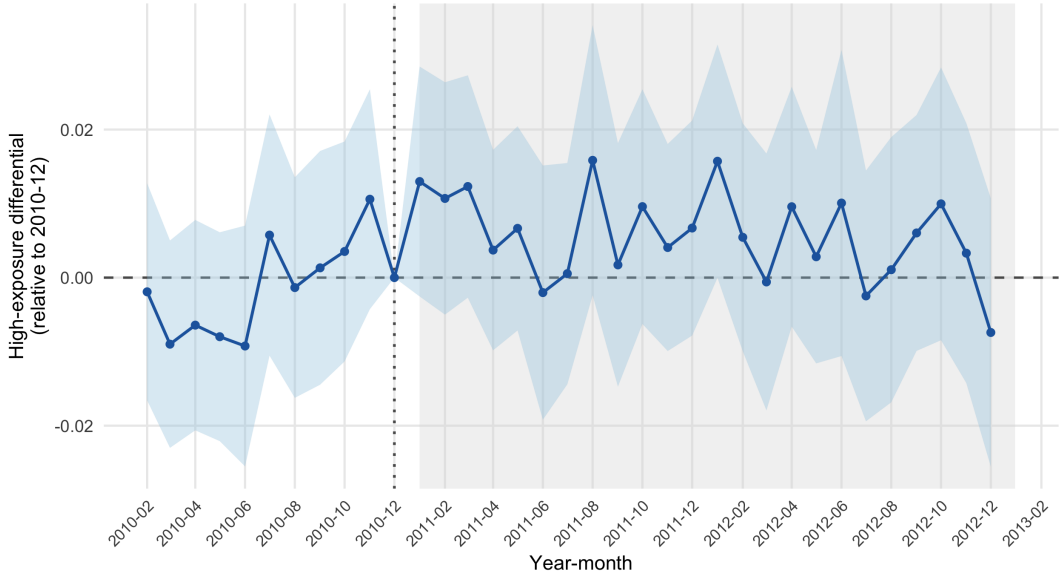



Figure 9: Monthly Event-Study Coefficients: Tax-Exposure Design

*Notes:* The figure plots the month-specific coefficients on $1\{t = m\} \times \text{HighExposure}_i$ from the Fact 1 event-study specification. The regression includes individual fixed effects, year-month fixed effects, age fixed effects, and month-by-pre-claimer interactions. After estimation, the full series is explicitly normalized so that December 2010 equals zero, and every other month is plotted relative to that benchmark. The vertical line marks January 2011, the first month of the payroll-tax holiday.

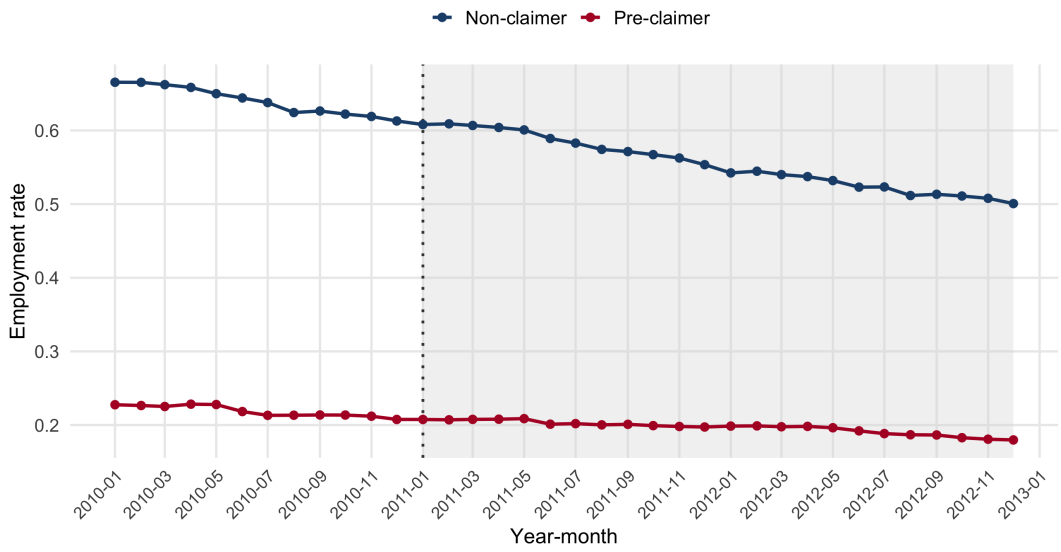



Figure 10: Monthly Group Trends: Claiming-Status Design

*Notes:* The figure plots monthly employment rates for pre-claimers and non-claimers in the sample used for Fact 2. The vertical line marks January 2011, the first month of the payroll-tax holiday. These are descriptive monthly paths; the main regression adds person fixed effects, year-month fixed effects, and the survival interactions discussed in the text.

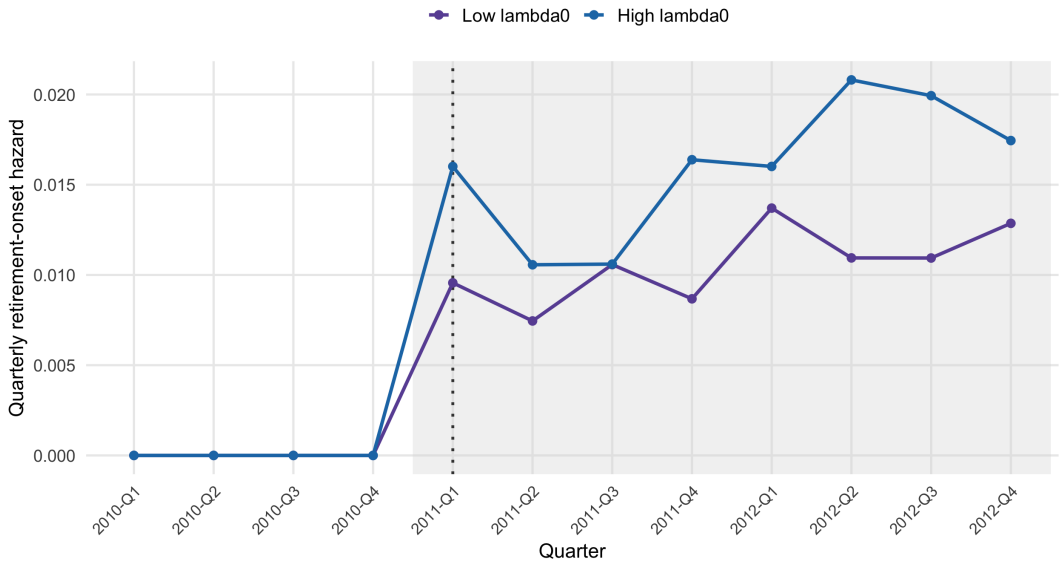



Figure 11: Quarterly Group Trends: High-$\lambda_0$ vs. Low-$\lambda_0$ Among Unclaimed Workers

*Notes:* The figure plots quarterly retirement-onset hazards for unclaimed workers above and below the median baseline $\lambda_0$. The vertical line marks 2011:Q1, the first quarter of the payroll-tax holiday. The main specification in Fact 3 uses continuous $\lambda_0$, so this median split is a descriptive companion rather than the estimating equation itself.

## B.2 Aggregate Retirement Claiming: SSA Administrative Evidence

Table 8 reports the number of retired-worker awards (new claims for Social Security retired-worker benefits) from SSA administrative records (Annual Statistical Supplement, Table 6.B5) for years 2005–2017. The table shows annual totals (in thousands) and the percentage of awards at age 62, separately by sex.

Table 8: Retired-Worker Awards by Year, 2005–2017 (SSA Table 6.B5)

| | Awards (thousands) | | | Average age at award | | | % claiming at 62 | |
|---|---|---|---|---|---|---|---|---|
| Year | Men | Women | Total | Men | Women | All | Men | Women |
| 2005 | 1,061 | 939 | 2,000 | 63.6 | 63.6 | 63.6 | 50.2 | 54.3 |
| 2006 | 1,057 | 942 | 1,999 | 63.8 | 63.8 | 63.8 | 45.6 | 50.2 |
| 2007 | 1,075 | 961 | 2,036 | 64.0 | 63.9 | 64.0 | 42.4 | 47.6 |
| 2008 | 1,200 | 1,079 | 2,279 | 64.0 | 63.9 | 64.0 | 42.5 | 48.3 |
| 2009 | 1,452 | 1,288 | 2,740 | 64.0 | 63.9 | 64.0 | 44.3 | 49.8 |
| 2010 | 1,387 | 1,248 | 2,635 | 64.1 | 63.9 | 64.0 | 42.6 | 47.9 |
| 2011 | 1,340 | 1,238 | 2,578 | 64.0 | 63.8 | 63.9 | 41.4 | 46.5 |
| 2012 | 1,419 | 1,316 | 2,735 | 64.2 | 64.0 | 64.1 | 37.2 | 42.4 |
| 2013 | 1,447 | 1,347 | 2,794 | 64.3 | 64.2 | 64.3 | 34.8 | 40.2 |
| 2014 | 1,433 | 1,339 | 2,772 | 64.4 | 64.2 | 64.3 | 34.5 | 39.7 |
| 2015 | 1,475 | 1,364 | 2,839 | 64.5 | 64.4 | 64.5 | 31.9 | 37.4 |
| 2016 | 1,519 | 1,392 | 2,911 | 64.6 | 64.5 | 64.6 | 30.4 | 35.5 |
| 2017 | 1,531 | 1,444 | 2,975 | 64.7 | 64.6 | 64.7 | 29.0 | 32.9 |

Source: Social Security Administration, Annual Statistical Supplement, Table 6.B5 (Master Beneficiary Record, 100% data). Blue-shaded rows mark the payroll tax holiday period (2011–2012). Gray-shaded row marks the 2009 recession spike. "Awards" are new retired-worker benefit claims by calendar year of award action. "Average age" and "% claiming at 62" are computed from the age distribution reported in the same table. The "All" average age column is the unweighted mean of the sex-specific averages.

## B.3 Inference and Sample-Composition Robustness

# C Earnings Proxy Validation

The calibration in Section 6.4.1 assigns workers to PIA brackets using current or recent earnings observed in the SIPP, under the assumption that current earnings approximate a worker's position in the Social Security benefit formula. A potential concern is that current earnings may not accurately reflect lifetime earnings position ($\mathrm{AIME}$): a high-wealth senior working part-time could have high lifetime earnings but low current earnings, leading to misclassification of the marginal replacement rate $r(e)$ and, consequently, the benefit accrual rate $\lambda_0(e)$.

This appendix uses administrative Social Security benefit data linked to HRS respondents to assess the validity of this earnings proxy.

**Data.** The RAND HRS Social Security Wealth file (`SSWEALTH`) contains SSA-computed present discounted values of projected benefit streams at ages 62, FRA, and 70, derived from administrative earnings records. We extract the wave-10 (year 2010) benefit wealth at FRA (`R10SSWRNR`), which we convert to an approximate monthly PIA using age-specific actuarial annuity factors.[4] From the RAND HRS core file, we extract wave-10 earnings (`r10iearn`), work status, hours, and household wealth.

We merge these files on `hhidpn` and restrict to a *broad sample* of working adults aged 60–69 with valid earnings and valid SSA benefit wealth ($N = 1{,}186$), and a *near-cap sample* further restricted to earnings within $0.8$–$1.2$ times the 2010 taxable maximum of $\$106{,}800$ ($N = 99$).

**Bracket classification.** We classify each worker into a marginal PIA bracket by two methods. The *SSW-implied bracket* converts SSA benefit wealth at FRA to approximate monthly PIA and maps it to the 90%, 32%, or 15% bracket using the 2010 bend points ($b_1 = \$761$, $b_2 = \$4{,}586$) and the corresponding PIA kink thresholds ($\$684.90$ and $\$1{,}908.90$ per month). The *earnings-implied bracket* approximates monthly $\mathrm{AIME}$ as current annual earnings divided by 12 and maps it directly to the same bend points.

**Results.** Table 9 summarizes the five validation checks.

[4] We use a baseline annuity factor of 12.7 at FRA (age 66, 2.9% real discount rate, SSA period life table for males), adjusted for each respondent's age at the time of interview: for ages below FRA, the factor is discounted by $(1+r)^{-(66-a)}$; for ages above FRA, the factor is reduced by 0.85 per year past FRA to approximate the shorter remaining benefit horizon. The annuity factor ranges from 10.7 (age 60) to 12.7 (age 66) in our sample.

Table 9: Current Earnings as a Proxy for PIA Bracket Position: HRS Validation

| Check | Statistic | Value |
|---|---|---|
| *Panel A: Rank correlation* | | |
| Broad sample ($N = 1{,}186$) | Spearman $\rho$ (earnings vs. SSA wealth) | 0.516 |
| | Pearson $r$ | 0.498 |
| Near-cap sample ($N = 99$) | Spearman $\rho$ | 0.059 |
| *Panel B: Bracket classification accuracy* | | |
| Broad sample | Overall accuracy | 64.1% |
| | 15% bracket correct (of 385 classified) | 47.8% |
| | 32% bracket correct (of 676 classified) | 80.5% |
| Near-cap sample | Fraction in 15% bracket (SSW-implied) | 58.6% |
| *Panel C: Near-cap percentile check (annuity-factor-free)* | | |
| Near-cap workers in top 20% of SSW distribution | | 51.5% |
| Near-cap workers in top 40% of SSW distribution | | 69.7% |
| *Panel D: Full-time vs. part-time accuracy* | | |
| Full-time ($\geq 35$ hrs/wk, $N = 838$) | Bracket accuracy | 65.5% |
| Part-time ($< 35$ hrs/wk, $N = 317$) | Bracket accuracy | 62.2% |
| *Panel E: Wealth of misclassified workers* | | |
| Correctly classified ($N = 760$) | Median household wealth | $181,500 |
| Misclassified ($N = 426$) | Median household wealth | $202,310 |

*Notes:* Sample is HRS wave-10 (2010) male respondents aged 60–69 who are currently working with valid earnings and valid SSA benefit wealth from the `SSWEALTH` file. Earnings, SSA benefit wealth, and household wealth are winsorized at the 5th and 95th percentiles. SSA benefit wealth at FRA is converted to approximate monthly PIA using age-specific actuarial annuity factors (range 10.7–12.7). Bracket classification uses 2010 PIA bend points ($b_1 = \$761$, $b_2 = \$4{,}586$). The near-cap sample restricts to earnings within 0.8–1.2 times the taxable maximum ($\$106{,}800$). Panel C uses percentile ranks of SSA benefit wealth, which do not depend on the annuity factor conversion. Full-time is defined as $\geq 35$ hours per week at the primary job. Household wealth is total non-housing financial wealth (`h10atotb`).

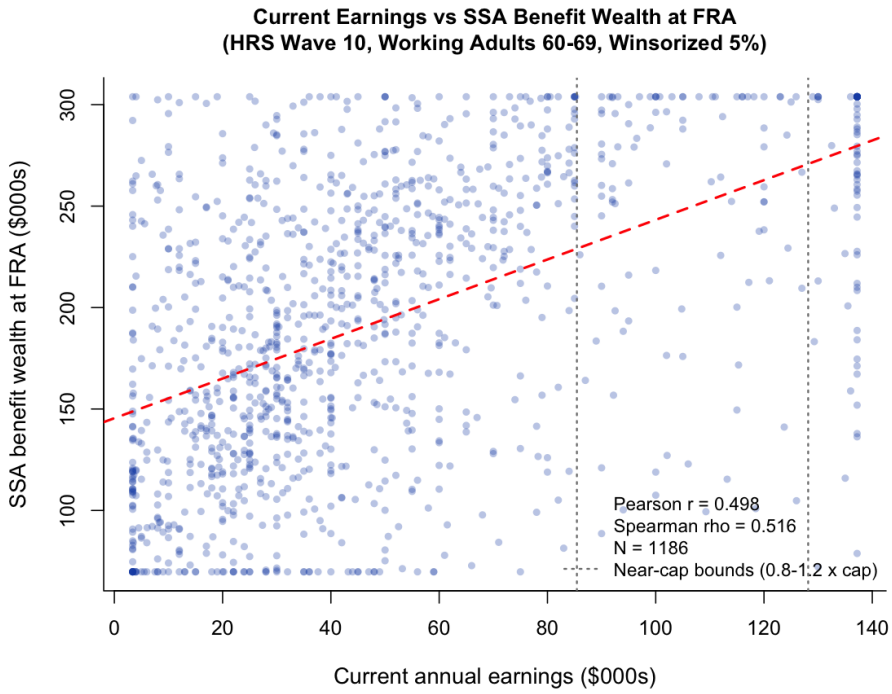



Figure 12: Current Earnings vs. SSA Benefit Wealth at FRA (HRS Wave 10)

*Notes:* Each point is a male HRS respondent aged 60–69 who is currently working, with both valid current earnings and valid SSA-linked benefit wealth at FRA from the `SSWEALTH` file ($N = 1{,}186$). Both variables are winsorized at the 5th and 95th percentiles. The dashed red line is the OLS fit. Vertical gray lines mark the near-cap region (0.8–1.2 times the 2010 taxable maximum of $106,800). Spearman $\rho = 0.516$; Pearson $r = 0.498$.

**Discussion.** Three findings bear on the credibility of the earnings-based bracket assignment.

*First,* the broad-sample Spearman rank correlation between current earnings and SSA benefit wealth is 0.516, indicating that current earnings preserve roughly half of the ordinal ranking in lifetime benefit position. The Pearson correlation is 0.498 after winsorizing at the 5th and 95th percentiles to limit the influence of extreme outliers.

*Second,* and most relevant for the paper's identification, near-cap workers are strongly concentrated in the upper tail of the SSA benefit wealth distribution: 51.5% are in the top quintile and 69.7% are in the top 40%. This percentile-based check does not depend on the annuity factor used to convert SSW to PIA, and confirms that workers earning near the taxable maximum do tend to have high lifetime earnings. When bracket classification is used, 58.6% of near-cap workers are assigned to the 15% bracket from SSA data, compared with 100% from earnings data; the remaining 41.4% are classified as 32% bracket, suggesting some measurement noise

in the SSW-to-PIA conversion rather than genuine misclassification, since the percentile evidence shows these workers are overwhelmingly in the upper tail.

*Third*, the part-time concern is modest. Bracket accuracy among part-time workers (62.2%) is only slightly lower than among full-time workers (65.5%), and misclassified workers have only modestly higher median household wealth ($202,310 vs. $181,500), weakly consistent with the "wealthy part-timer" mechanism but not of a magnitude that would substantially bias the main estimates.

Overall, the validation supports using current earnings as a proxy for PIA bracket position, particularly in the near-cap sample that drives the third empirical result. The rank correlation is moderate, the near-cap percentile concentration is strong, and the full-time/part-time differential is small.

# D Calibration Tables and Figures

This appendix collects all tables and figures referenced in Section 7.

Table 10: Targeted Moments

| Block | Moments | Count | Block weight |
|---|---|---|---|
| Age profile | Employment by age, 62–70 | 9 | 2.0 |
| Health profile | Employment by health group, ages 62–70 | 4 | 1.0 |

*Notes:* Within the age block, ages 62–65 receive higher weight than ages 66–70.

Table 11: Fixed Parameters

| Symbol | Parameter | Value |
|---|---|---|
| *Panel A: Standard or Normalized Parameters* | | |
| $\beta$ | Discount factor | 0.98 |
| $b$ | Asset grid | $(0.0,\ 1.5,\ 5.0)$ |
| $\omega_b$ | Asset weights | $(0.35,\ 0.40,\ 0.25)$ |
| $\nu$ | Work-preference support | $(-1,\ 0,\ 1)$ |
| $\omega_\nu$ | Work-preference weights | $(0.25,\ 0.50,\ 0.25)$ |
| $\tau_h, \tau_b$ | Health and asset wage tilts | 0.60, 0.45 |
| $\zeta_z, \rho_z$ | Bad-shock wage scale and persistence | 0.60, 0.75 |
| $s_x$ | Bad-shock exit scale | 0.35 |
| $\kappa_{62:65}$ | Early work-cost profile | $(0.15,\ 0.12,\ 0.08,\ 0.05)$ |
| *Panel B: Policy Parameters* | | |
| $a^*$ | Full Retirement Age | 66 |
| $\delta_D$ | Delayed retirement credit (per year) | 0.08 |
| $\theta, \theta^{FRA}$ | RET withholding rates | 0.50, 0.333 |
| $\bar{y}, \bar{y}^{FRA}$ | RET exempt earnings | \$14,160, \$37,680 |
| $\bar{w}$ | Taxable maximum | \$106,800 |
| $\tau, \tau'$ | Benchmark / counterfactual tax rates | 0.062, 0.042 |
| *Panel C: Author-Constructed from Data* | | |
| $\zeta_w$ | Wage-type multipliers | $(0.70,\ 1.00,\ 1.35)$ |
| $\omega_w$ | Wage-type weights | $(0.35,\ 0.45,\ 0.20)$ |
| $r^z_{H1:H4}$ | Health-shock arrival profile | $(1.000,\ 0.162,\ 0.110,\ 0.060)$ |

*Notes:* Panel A lists values fixed prior to SMM because they are either standard annual normalizations or reduced-form supports that are not separately identified by the targeted moments. We set $\beta = 0.98$ in line with annual retirement calibrations in French (2005); French and Jones (2011); Gustman and Steinmeier (2005). The asset grid is a reduced-form liquidity support because the simplified model has no savings choice and the current estimation sample does not contain a comparable wealth measure. The early work-cost profile is fixed to capture the steep decline in employment at ages 62–65 without adding more free parameters. Panel B reports statutory values from 2010 law. Panel C reports parameters we construct directly from the data. The wage support is a coarse three-point approximation to the 2010 SIPP earnings distribution of employed seniors: weighted annual earnings have mean \$41.9k, median \$27.2k, and 75th percentile \$52.6k, so the support $(\$21k, \$30k, \$40.5k)$ spans the center of the distribution while keeping the state space small. The health-shock profile is anchored to the quartile-specific survival ordering from the HRS-based survival mapping.

Table 12: SMM-Estimated Parameters

| Symbol | Parameter | Estimate |
|---|---|---|
| $\psi_0$ | Retirement-utility intercept | 0.9467 |
| $\psi_1$ | Age slope of work disutility | 0.0532 |
| $\psi_2$ | Post-65 work-disutility increment | 0.0067 |
| $\kappa_0$ | Fixed cost of work | 0.5957 |
| $\chi$ | Cost of remaining unclaimed | 0.4118 |
| $q$ | Bad-shock arrival probability | 0.1326 |
| $p_f$ | Bad-shock exit intensity | 0.4834 |
| $\kappa_h$ | Health-work disutility scale | 0.2751 |
| $r_{H2}$ | Health-work rank, group 2 | 0.1309 |
| $r_{H3}$ | Health-work rank, group 3 | 0.0848 |
| $r_{H4}$ | Health-work rank, best health | 0.0435 |

*Notes:* The worst-health rank is normalized to one, so only $(r_{H2}, r_{H3}, r_{H4})$ are estimated. These parameters are jointly estimated by SMM from the 13 targeted employment moments.

Table 13: Fit to Targeted Moments

| *Employment by Age* | | | | *Employment by Health Group, 62–70* | | | |
|---|---|---|---|---|---|---|---|
| Age | Data | Model | Diff. | Group | Data | Model | Diff. |
| 62 | 0.512 | 0.528 | +0.016 | $H1$ | 0.152 | 0.127 | −0.026 |
| 63 | 0.461 | 0.473 | +0.012 | $H2$ | 0.381 | 0.342 | −0.039 |
| 64 | 0.426 | 0.433 | +0.008 | $H3$ | 0.446 | 0.397 | −0.050 |
| 65 | 0.351 | 0.379 | +0.027 | $H4$ | 0.477 | 0.447 | −0.030 |
| 66 | 0.306 | 0.345 | +0.039 | | | *RMSE* | *0.037* |
| 67 | 0.272 | 0.319 | +0.046 | | | | |
| 68 | 0.247 | 0.281 | +0.034 | | | | |
| 69 | 0.230 | 0.241 | +0.011 | | | | |
| 70 | 0.207 | 0.200 | −0.007 | | | | |
| | | *RMSE* | *0.026* | | | | |

*Notes:* The SMM criterion targets all 13 moments. The final weighted objective value is 0.00246. $H1$ denotes the worst health and $H4$ denotes the best health case.

Table 14: Permanent Payroll-Tax Cut: Employment Responses by Group

| Margin | Group | Benchmark | Counterfactual | Change |
|---|---|---|---|---|
| Overall | Ages 62–70 | 0.358 | 0.370 | +0.012 |
| Age | 62–65 | 0.453 | 0.464 | +0.011 |
| Age | 66–70 | 0.277 | 0.290 | +0.013 |
| Health | $H1$ (worst) | 0.127 | 0.127 | +0.000 |
| Health | $H2$ | 0.342 | 0.349 | +0.007 |
| Health | $H3$ | 0.397 | 0.410 | +0.013 |
| Health | $H4$ (best) | 0.447 | 0.465 | +0.019 |
| Assets | Low | 0.302 | 0.317 | +0.015 |
| Assets | Middle | 0.348 | 0.359 | +0.011 |
| Assets | High | 0.452 | 0.463 | +0.011 |
| Earnings | \$30k | 0.355 | 0.368 | +0.012 |
| Earnings | \$60k | 0.361 | 0.367 | +0.006 |
| Earnings | \$90k | 0.361 | 0.365 | +0.003 |
| Earnings | \$110k | 0.355 | 0.359 | +0.004 |
| Earnings | \$150k | 0.350 | 0.355 | +0.005 |

*Notes:* The overall row uses the benchmark age distribution over ages 62–70. Earnings rows come from auxiliary type-by-type counterfactuals holding all other calibrated parameters fixed.

Table 15: Channel Decomposition of the Employment Response

*Panel A: By asset group*

| Group | Total | Baseline | Waiting-cost | Wage-sorting | Interaction |
|---|---|---|---|---|---|
| Low | 1.458 | 2.092 | +0.167 | −0.771 | −0.029 |
| Middle | 1.076 | 2.092 | −1.016 | 0.000 | 0.000 |
| High | 1.081 | 2.092 | −0.737 | +0.895 | −1.169 |

*Panel B: By health group*

| Group | Total | Baseline | Longevity | Work-capacity | Wage-sorting |
|---|---|---|---|---|---|
| $H1$ | 0.041 | 0.608 | −0.436 | −0.130 | −0.001 |
| $H2$ | 0.683 | 0.608 | +0.100 | −0.101 | +0.075 |
| $H3$ | 1.343 | 0.608 | −0.090 | +0.642 | +0.183 |
| $H4$ | 1.859 | 0.608 | +0.107 | +0.491 | +0.653 |

*Notes:* Entries are percentage-point changes in pooled employment, ages 62–70. In each panel, "Baseline" is the response when all panel-specific channels are shut off. Panel A channels: "Waiting-cost" is the extra contribution of asset-dependent claiming costs; "Wage-sorting" is the extra contribution of the asset tilt in wage-type composition; "Interaction" is the residual. Panel B channels: "Longevity" adds back health-specific survival differences; "Work-capacity" adds back health-related work disutility and bad-shock risk; "Wage-sorting" adds back the health tilt in wage-type composition.

## References


Behaghel, L. and Blau, D. M. (2012). Framing social security reform: Behavioral responses to changes in the full retirement age. *American Economic Journal: Economic Policy*, 4(4):41–67.

Blundell, R., French, E., and Tetlow, G. (2016). Retirement incentives and labor supply. In Piggott, J. and Woodland, A., editors, *Handbook of the Economics of Population Aging*, volume 1, pages 457–566. Elsevier.

Chetty, R., Guren, A., Manoli, D., and Weber, A. (2013). Does indivisible labor explain the difference between micro and macro elasticities? a meta-analysis of extensive margin elasticities. *NBER Macroeconomics Annual*, 27:1–56.

Coile, C. and Gruber, J. (2007). Future social security entitlements and the retirement decision. *The Review of Economics and Statistics*, 89(2):234–246.

Coile, C. C. (2023). Changing retirement incentives and retirement in the US. In *Social Security Programs and Retirement around the World: The Effects of Reforms on Retirement Behavior*, chapter 12. University of Chicago Press.

Cruz, M., Covinsky, K., Widera, E. W., Stijacic-Cenzer, I., and Lee, S. J. (2013). Predicting 10-year mortality for older adults. *JAMA*, 309(9):874–876.

De Nardi, M., Fella, G., and Paz-Pardo, G. (2016). The implications of richer earnings dynamics for consumption and wealth. NBER Working Paper 21917, National Bureau of Economic Research.

De Nardi, M., French, E., and Jones, J. B. (2006). Differential mortality, uncertain medical expenses, and the saving of elderly singles. NBER Working Paper 12554, National Bureau of Economic Research.

Feldstein, M. (2005). Rethinking social insurance. *American Economic Review*, 95(1):1–24.

Feldstein, M. and Samwick, A. A. (1992). Social security rules and marginal tax rates. *National Tax Journal*, 45(1):1–22.

Finkelstein, A. (2007). The aggregate effects of health insurance: Evidence from the introduction of medicare. *The Quarterly Journal of Economics*, 122(1):1–37.

French, E. (2005). The effects of health, wealth, and wages on labour supply and retirement behaviour. *The Review of Economic Studies*, 72(2):395–427.

French, E. and Jones, J. B. (2011). The effects of health insurance and self-insurance on retirement behavior. *Econometrica*, 79(3):693–732.

Gelber, A. M., Jones, D., Sacks, D. W., and Song, J. (2022). The employment effects of the social security earnings test. *Journal of Human Resources*, 57(2):341–371.

Goss, S. C. (2019). Testimony by stephen c. goss, chief actuary, social security administration, before the house committee on ways and means. SSA testimony. July 25, 2019.

Gustman, A. L. and Steinmeier, T. L. (1986). A structural retirement model. *Econometrica*, 54(3):555–584.

Gustman, A. L. and Steinmeier, T. L. (2005). The social security early entitlement age in a structural model of retirement and wealth. *Journal of Public Economics*, 89(2–3):441–463.

Hurd, M. D. and McGarry, K. (1995). Evaluation of the subjective probabilities of survival in the health and retirement study. *Journal of Human Resources*, 30:S268–S292.

Hurd, M. D. and McGarry, K. (2002). The predictive validity of subjective probabilities of survival. *The Economic Journal*, 112(482):966–985.

Hurd, M. D., Smith, J. P., and Zissimopoulos, J. M. (2004). The effects of subjective survival on retirement and social security claiming. *Journal of Applied Econometrics*, 19(6):761–775.

Krueger, A. B. and Pischke, J.-S. (1992). The effect of social security on labor supply: A cohort analysis of the notch generation. *Journal of Labor Economics*, 10(4):412–437.

Laun, T. and Wallenius, J. (2016). Social insurance and retirement: A cross-country perspective. *Review of Economic Dynamics*, 22:72–92.

Lee, S. J., Lindquist, K., Segal, M. R., and Covinsky, K. E. (2006). Development and validation of a prognostic index for 4-year mortality in older adults. *JAMA*, 295(7):801–808.

Liebman, J. B. and Luttmer, E. F. P. (2015). Would people behave differently if they better understood social security? evidence from a field experiment. *American Economic Journal: Economic Policy*, 7(1):275–299.

Liebman, J. B., Luttmer, E. F. P., and Seif, D. G. (2009). Labor supply responses to marginal social security benefits: Evidence from discontinuities. *Journal of Public Economics*, 93(11–12):1208–1223.

Mastrobuoni, G. (2009). Labor supply effects of the recent social security benefit cuts: Empirical estimates using cohort discontinuities. *Journal of Public Economics*, 93(11–12):1224–1233.

Ohanian, L., Raffo, A., and Rogerson, R. (2008). Long-term changes in labor supply and taxes: Evidence from OECD countries, 1956–2004. *Journal of Monetary Economics*, 55(8):1353–1362.

Prescott, E. C. (2004). Why do Americans work so much more than Europeans? *Federal Reserve Bank of Minneapolis Quarterly Review*, 28(1):2–13.

Rogerson, R. and Wallenius, J. (2009). Micro and macro elasticities in a life cycle model with taxes. *Journal of Economic Theory*, 144(6):2277–2292.

Rust, J. and Phelan, C. (1997). How social security and medicare affect retirement behavior in a world of incomplete markets. *Econometrica*, 65(4):781–831.

Social Security Administration, Office of the Chief Actuary (2025). Summary of provisions that would change the social security program: Provisions affecting payroll taxes. SSA OCACT policy options page. Estimates based on the 2025 Trustees Report.

Steingrimsson, J. A., Gatsonis, C., Li, B., and Dahabreh, I. J. (2023). Transporting a prediction model for use in a new target population. *American Journal of Epidemiology*, 192(2):296–304.

Stock, J. H. and Wise, D. A. (1990). Pensions, the option value of work, and retirement. *Econometrica*, 58(5):1151–1180.

Yaari, M. E. (1965). Uncertain lifetime, life insurance, and the theory of the consumer. *The Review of Economic Studies*, 32(2):137–150.